



QF5206: Group Project Report

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Report

Research Background

Against the backdrop of an increasingly complex global geopolitical landscape, frequent local conflicts, and trade frictions, the energy market, as a core national strategic resource and a key component of macroeconomic operations, faces intense price volatility due to strong impacts from non-market factors such as policy interventions, geopolitical events, and supply chain disruptions. Compared to traditional financial assets like stocks and bonds, energy assets (such as crude oil, natural gas, coking coal, and coke) exhibit higher volatility, more significant policy sensitivity, and more complex non-linear characteristics. The price series of these assets often deviate from the assumptions of stationarity and normal distribution found in classic financial models, presenting features like leptokurtosis (sharp peaks and fat tails) and volatility clustering, which pose severe challenges for prediction and modeling.

In this highly uncertain market environment, identifying asset combinations (trading pairs) with stable statistical relationships has become a crucial pathway to controlling risk and securing stable returns. The core value of pairs trading strategies lies in constructing investment portfolios with controllable risk exposure by leveraging the intrinsic correlations between assets.

Trading Pairs

Intrinsic Correlation Mechanisms

Positively Correlated Pairs (e.g., WTI Crude vs. Brent Crude) These assets are driven by similar macroeconomic factors, resulting in highly synchronized price movements. The strategic value lies in capturing persistent trends in their relative strength or in stripping out unnecessary directional risk by constructing a hedged portfolio.

Supply Chain Transmission Pairs (e.g., Crude Oil vs. Refined Products) By analyzing the price transmission efficiency and temporal relationships between upstream and downstream segments of the supply chain, momentum strategies based on differences in transmission speed can be developed.

Substitution Pairs (e.g., Natural Gas vs. Coal) These assets have substitutable economic properties. Changes in their price ratio often signal shifts in energy consumption structure, allowing for the development of trend strategies based on threshold breaches.

Risk Hedging Pairs (e.g., Crude Oil vs. New Energy Stocks) By leveraging the different responses of traditional energy and emerging energy assets to the same risk factor (such as environmental policies), portfolios can be constructed to hedge specific risk factors.

Typical Energy Market Trading Pairs and Their Inner Logic

Crack Spread This reflects crude oil refining margins, connecting crude oil with refined products (like gasoline, diesel). The spread is jointly affected by refinery operating rates, seasonal demand (summer driving season, winter heating season), inventory levels, and environmental policies. When the spread deviates from its reasonable range, it signifies abnormal refining profits, and the market will drive the spread back to its mean by adjusting refinery capacity or product supply.

Coking Coal–Coke Spread This embodies the cost-profit relationship between raw materials and finished products in the steel industry chain. Coking coal is processed into coke, which is used for blast furnace ironmaking. This spread is significantly influenced by steel production capacity, environmental production-limit policies, and coal supply. Its fluctuation reflects the profit distribution structure within the ferrous metal industry chain.

Cross-Regional Spreads (e.g., WTI vs. Brent Crude) This reflects supply-demand differences, transportation costs, and geopolitical risks between different markets. The widening or narrowing of the spread between two locations is often driven by factors like pipeline capacity, export policies, and regional conflicts, providing opportunities for cross-market arbitrage.

Energy Substitution Pairs (e.g., Natural Gas vs. Coal) In the power generation sector, natural gas and coal, as primary fuels, have a clear substitution effect. Changes in their price ratio directly influence the fuel choices of power generation companies, thus creating cross-commodity arbitrage opportunities.

New Energy Industry Chain Pairs (e.g., Lithium Miner Stocks vs. Battery Manufacturers) This reflects the profit transmission between the upstream and downstream segments of the new energy industry chain. A lagged but stable relationship exists between the prices of upstream resources and the stock prices of midstream manufacturing companies, which can be used to capture relative value opportunities arising from industrial cycle fluctuations.

Biofuel Substitute Pairs (e.g., Soybean Oil and Palm Oil) Similar to natural gas and coal, there is also a significant substitution effect between biofuels. Consumer demand shifts with price fluctuations, which helps to stabilize the price relationship.

Trading Strategies

Spread Mean Reversion

In the commodity futures market, spread arbitrage strategies occupy an important position and have become a key component of quantitative trading because they offer significant advantages over single-sided speculative trading. The core of these advantages is that spread arbitrage can effectively strip out unnecessary market volatility, focusing on the deterministic patterns of relative price relationships to achieve stable returns at lower risk.

First, the core advantage of spread arbitrage lies in its ability to capitalize on the mean-reverting property of spreads. The price difference between different contracts of the same commodity or between highly correlated commodities is usually constrained within a relatively limited range, tending to fluctuate around a long-term equilibrium value due to a shared macroeconomic background and the commodities' intrinsic values. When the spread excessively deviates from its historical mean due to short-term market sentiment, capital disturbances, or liquidity shocks, an arbitrage opportunity arises. Traders build positions in opposite directions, betting on the spread's reversion to the mean. The profit basis lies in correcting this short-term mispricing, rather than predicting the market's absolute direction. This mean-reverting characteristic makes the spread's movement more predictable than the absolute price movement of a single contract.

Second, spread trading has a natural risk-hedging effect, effectively stripping out most of the impact from macroeconomic systemic risks. When a trader establishes both long and short positions simultaneously, systemic risk factors that affect both contract prices in the same direction—such as interest rate changes, macroeconomic cycles, and geopolitical events—are largely offset. This ensures that the value change of the arbitrage portfolio depends mainly on the relative strength between the two contracts, i.e., differences in micro-fundamental factors, such as the supply-demand situation of a specific contract, seasonal patterns, storage costs, or policy impacts. By focusing on these relatively controllable, fundamentally-based factors, spread arbitrage reduces reliance on unpredictable macroeconomic volatility.

Furthermore, from a risk-return perspective, spread arbitrage typically exhibits lower volatility and limited risk. By hedging directional risk, the net value fluctuation of a spread portfolio is far less than that of a single-sided position. This low-volatility characteristic means that in extreme market conditions, even if prices hit the daily up or down limit, the loss on an arbitrage position is often far smaller than in single-sided speculation, as the loss on one leg is partially or fully offset by the profit on the other. Consequently, spread arbitrage offers a more attractive risk-return ratio, which is more conducive to the stable operation of large capital.

Finally, from the perspectives of trading psychology and capital utilization efficiency, spread arbitrage also has advantages. Its hedging mechanism smooths the portfolio's equity curve, alleviating the psychological pressure on traders from severe market fluctuations and contributing to long-term decision-making stability. Meanwhile, because the risk is relatively controllable, traders can deploy higher leverage in a single trade, thus enhancing the time-based efficiency of capital utilization.

In summary, spread arbitrage achieves the goal of capturing opportunities with higher certainty in complex market environments by focusing on mean reversion, stripping out macro noise, and hedging directional risk. This is the fundamental reason why it has endured in the field of quantitative investment and has become an ideal application scenario for advanced technologies like machine learning.

Cointegration and Error Correction Model

The Cointegration and Error Correction Model (ECM) serves as the statistical foundation for pairs trading and multi-variety arbitrage strategies. It transcends the simple spread (e.g., $P_1 - P_2$) mean reversion assumption, aiming to capture the long-term equilibrium relationship between non-stationary time series (i.e., series whose mean and variance change over time, like most commodity or stock prices).

First, cointegration tests (such as the Engle-Granger two-step method or the Johansen test) are used to determine if a linear combination exists between two or more non-stationary series (e.g., commodity futures prices P_1 and P_2) such that the resulting combined series (the residual or equilibrium error) is stationary (i.e., its mean and variance do not change over time, and the mean reverts around a constant). After confirming cointegration, the Error Correction Model (ECM) is used to model the short-term dynamic adjustment process. It links the short-term changes in the series ($\Delta P_{1,t}$ and $\Delta P_{2,t}$) with the previous period's equilibrium error (S_{t-1}).

The core logic of the model is: when the short-term spread deviates from the long-term equilibrium ($S_{t-1} \neq 0$), a corrective force is generated, pushing them back towards long-term equilibrium. The ECM captures the speed and direction of this corrective force. Trading signals are based on the deviation degree of the equilibrium error S_t (e.g., standard deviation multiples σ from its mean μ). When S_t breaches a threshold (e.g., $\mu + 2\sigma$), it indicates excessive short-term deviation, and the model predicts a correction towards μ , at which point a reverse arbitrage operation is executed.

Compared to simple spreads, ECM is based on rigorous time-series statistics (cointegration theory) and can more accurately identify causal relationships and dynamic adjustment mechanisms between prices, enhancing the statistical arbitrage nature of the strategy. The strategy captures short-term imbalances deviating from long-term equilibrium, rather than just spread volatility, making the trading signals more representative of actual arbitrage opportunities and helping to avoid the "spurious regression" trap. Cointegration testing effectively filters out trading pairs that lack a long-term constraint. If the spread itself is non-stationary, any attempt at mean reversion could lead to losses. The premise of ECM ensures the stationarity of the residual series, fundamentally reducing this risk. ECM can also provide a basis for dynamic position adjustments. The model parameters can be used to estimate the speed of reversion, which helps optimize take-profit and stop-loss settings.

Fundamentals-Driven Relationship Strength Assessment

The core of this strategy lies in combining traditional technical spread mean reversion with macroeconomic or industry-specific fundamental analysis to achieve dynamic position management. This is because the relationship strength between a trading pair is not constant; it fluctuates with changes in driving factors (such as supply-demand relationships, policy environments, and macroeconomic cycles).

First, for a specific trading pair, identify the core fundamental factors affecting their relative prices. Then, transform these factors into quantifiable and trackable indicators, for example: inventory ratios (excessively high or low inventories can affect the stability of the refined products vs. crude oil spread), macroeconomic sentiment indicators (like PMI, credit impulse), etc. Next, use rolling regression, Kalman filters, or other state-space models to estimate the cointegration parameters between the pair (such as the hedge ratio β) and the residual volatility (the spread's standard deviation σ). The constructed fundamental factors are introduced into the model as auxiliary variables (or external regressors) to predict or explain changes in β or σ . The constructed fundamental factors are introduced into the model as auxiliary variables (or external regressors) to predict or explain changes in β or σ .

In terms of position and risk management: when fundamental factors indicate the driving relationship is stable or strengthening (e.g., inventories are at a long-term equilibrium, policy expectations are stable), it implies an increased likelihood and speed of spread reversion. At this time, increase the arbitrage position size and trading frequency. When fundamental factors show the relationship is unstable or breaking down (e.g., volatile inventory fluctuations or significant policy uncertainty), it implies an increase in residual volatility σ and a higher risk of spread divergence. At this time, reduce the arbitrage position or close it entirely to decrease market exposure. σ and a higher risk of spread divergence. At this time, reduce the arbitrage position or close it entirely to decrease market exposure.

In summary, a fundamentals-based strategy can distinguish between spread deviations caused by random fluctuations and trend-like divergences caused by structural changes in fundamentals. By trading only when fundamentals support a stable relationship, the quality of trading signals is effectively improved. The fundamental factors act as risk indicators, providing early warnings before a relationship breaks down, guiding the strategy to proactively reduce positions and avoid significant losses during periods of systemic risk.

Multi-Period Mean Reversion Strategy

The multi-period mean reversion strategy recognizes that market price fluctuations are influenced by drivers on different time scales, including short-term trader sentiment, medium-term capital flows, and long-term fundamental value. The goal of this strategy is to isolate and superimpose the reversion opportunities from these different time scales, thereby enhancing signal robustness and capturing a richer set of return sources. This involves time scale division and model construction.

Specifically, the short-term cycle focuses on capturing instantaneous imbalances driven by market microstructure and liquidity. For example, using minute or hourly data to build autoregressive models (AR/ARMA) based on high-frequency spreads (or residuals) to capture rapid reversions after brief overbought/oversold conditions. The medium-term cycle focuses on capturing market sentiment fluctuations caused by macro events, capital flows, or news shocks. For example, using daily data to build traditional cointegration or simple spread mean reversion models to capture reversions to the mean after prices have deviated for several days or weeks. The long-term cycle focuses on capturing fundamental value reversion. For example, this might involve combining fundamental data or using regression models with long-term moving averages or exponential smoothing to capture value reversion on a monthly scale.

In implementation, signal decomposition techniques (such as wavelet analysis, Empirical Mode Decomposition (EMD), or simple multi-period residual separation) are used to break down the original spread into short-, medium-, and long-term components. When signals from multiple periods (especially short and medium) are aligned, it is considered a high-confidence trading opportunity, and position size can be increased. For example, a short-term oversold condition combined with a medium-term low deviation creates a combined force. When signals from different periods conflict (e.g., a short-term oversold signal suggests buying, but a long-term trend model shows the long-term mean is still declining), the strategy must make a decision based on preset weights or risk preferences.

Typically, the short-term signal submits to the medium/long-term one, or the conflicting signal is treated as low-confidence, and the position size is reduced. On each time scale, rolling regression or Kalman filters can be used to estimate independent periodic hedge ratios β_i . The final trading hedge ratio can be a weighted average of these periodic β_i , allowing for more flexible adaptation to different market states. The final trading hedge ratio can be a weighted average of these periodic β_i , allowing for more flexible adaptation to different market states.

Event-Driven Trading

Event-driven spread trading is a proactive arbitrage strategy that abandons the traditional "continuous reversion" assumption. Instead, it focuses on capitalizing on the short-term, predictable distortions created by exogenous shocks on specific trading pair spreads, betting that the spread will quickly recover to its pre-shock equilibrium level once the shock subsides. Examples include geopolitical events, policy announcements, extreme weather, or sudden data releases.

The core of the strategy is to assess whether the event's impact on the two assets in the pair is asymmetrical. For example, a hurricane hitting the Gulf of Mexico might have an immediate impact on the supply side of gasoline (RBOB) (refinery shutdowns), but a relatively delayed or smaller impact on crude oil (WTI). This asymmetric shock would cause the WTI-RBOB spread (i.e., refining margin) to widen significantly in the short term.

When the spread deviates excessively due to the shock (e.g., refining margins are pushed extremely high), the strategy establishes a counter-position. The strategy predicts that once the shock fades (e.g., hurricane warning lifted, policy impact digested), the market will reassess the fundamentals, and the spread will rapidly and forcefully revert to its historical equilibrium level. Trade execution requires extremely high speed and very low slippage, with entry often occurring after the event's impact has peaked.

Machine Learning-Based Non-Linear Relationship Capture

The machine learning-based non-linear relationship capture strategy aims to address the limitations of traditional statistical arbitrage models (like OLS linear regression, cointegration ECM) when dealing with complex, high-dimensional, and non-stationary financial data. It utilizes ML algorithms to identify complex patterns and non-linear dynamics in spread series that are difficult for humans to perceive, such as changes in reversion strength under volatility clustering or non-linear threshold effects of market sentiment. It can effectively integrate the technical features of the spread, market microstructure features, and quantitative fundamental factors into a unified predictive framework, maximizing information utilization.

ML models require rich input features to capture the complex drivers behind the spread. Besides traditional features like the spread itself, volatility, and volume, these often include: higher-moment features, such as the spread's skewness and kurtosis, to capture asymmetric reversion (reversion speed differs for positive and negative deviations); volatility clustering features, like the conditional volatility from GARCH models, to capture the impact of volatility clustering on reversion speed; interaction features, such as the product of asset A's volatility and asset B's return, to capture complex interactions; and market microstructure features, like order book depth and bid-ask spreads.

In terms of model selection, ensemble learning models like Random Forest are outstanding at capturing non-linear relationships, handling high-dimensional features, and identifying feature importance. They can be used to predict whether the spread will revert (classification) or the magnitude of the reversion (regression). Deep learning models like Recurrent Neural Networks (RNNs) are adept at processing time-series data, automatically learning and extracting temporal dependencies, making them particularly suitable for capturing the time-series patterns of spreads. State-space models can predict the transition matrices or observation noise in models like the Kalman filter, achieving non-linear, dynamic prediction of the hedge ratio β and residual volatility σ .

It is worth noting that ML models usually do not predict prices directly. Instead, they predict the probability of the spread reverting to its mean within a future time window (classification) or the target value/magnitude of the spread's reversion (regression). This can be directly used to guide dynamic position allocation. The higher the predicted probability, the larger the allocated position, enabling fine-grained, risk-adjusted management.

Strategy Used in This Project

In this project, we focus on building a systematic and extensible spread arbitrage framework, with energy assets as the primary research subject. Due to limitations in the research period and data availability, we selected three representative trading pairs as empirical cases: the coking coal–coke spread, the crude oil crack spread, and the soybean oil–palm oil spread. We conduct an in-depth study based on four modules: data processing, feature engineering, model training, and backtest evaluation. The spread mean reversion model we adopted is one of the most classic and robust methods in statistical arbitrage, possessing good interpretability and operability.

Although we did not cover all potential trading pairs or more complex strategic ideas, the framework we built has good generalization capabilities. It supports the integration of advanced methods like cointegration testing and multi-period analysis, providing a reliable technical platform for future expansion to other energy asset portfolios and more complex trading strategies. This research provides a methodological foundation and practical validation for systematically mining relative value opportunities in the highly uncertain energy market.

Project Overview

This project aims to build a modular and automated spread arbitrage framework, covering the entire workflow from data acquisition to strategy backtesting. By integrating data processing, feature engineering, machine learning model training, and backtest validation, it provides an efficient and extensible quantitative research tool.

Core Methodology

The project employs machine learning models (such as XGBoost, LightGBM) combined with fundamental relationships to predict commodity spreads, and validates the strategy's effectiveness through a backtesting module. The feature engineering module generates key features (such as historical spread volatility, seasonal factors) and enhances model performance through hyperparameter optimization.

Project Purpose

The spread arbitrage framework focuses on integrating the workflow from data acquisition to strategy validation. The division of four separate modules (data processing module, feature engineering module, model training module, and back-testing & evaluation module) allows flexible configuration of parameters under different hypotheses. Based on this framework, three typical commodity spread arbitrage strategies were modelled, including the crack spread, the coke-coking coal spread, and the soybean oil-palm oil spread. A good risk-return pattern is expected for subsequent strategy optimisation.

Furthermore, the project emphasises the generality and adaptability of the constructed tool, aiming to support the development of multiple commodity spread arbitrage strategies. By abstracting common operational interfaces, providing a diverse model library, and offering a backtest engine, the framework enables users to concentrate their research efforts on the strategy logic itself, rather than on the underlying technical implementation. The three typical strategy cases already implemented demonstrate that the framework is not only applicable to modeling classic spread combinations like coke-coking coal and crack spreads, but also possesses the potential to be extended to other related asset combinations. It thus provides a reliable technical platform for research in commodity futures spread trading.

Project Innovation

Compared to existing research, the primary innovations of this project include the following:

The first innovation is the adoption of a highly modular structure. The project deconstructs the entire spread strategy development life cycle into four distinct modules: data processing, feature engineering, model training, and backtest evaluation. These modules communicate through standardised interfaces, which both ensures procedural continuity and provides the flexibility for each component to be individually upgraded or replaced. This design not only reduces

system maintenance complexity but also enables researchers to conduct deep optimisation on specific stages—such as experimenting with different feature selection algorithms or ensemble models—without needing to refactor the entire analysis pipeline. This greatly enhances the scalability and reproducibility of the research.

Second, the project provides unified validation support for multiple categories of spread arbitrage strategies, moving beyond the limitations of typical research that focuses on a single, customised pair. By abstracting common data processing logic and model interfaces, the framework can flexibly adapt to the analytical requirements of different commodity combinations, such as coke-coking coal, crack spreads, and soybean oil-palm oil. This general-purpose design makes comparative strategy studies possible, allowing researchers to evaluate the stability and profitability of different spread strategies under a single, unified standard. This, in turn, provides a more reliable basis for asset allocation and improves the generalisability and persuasiveness of the research conclusions.

Achievements and Significance

In the case studies of coking coal-coke, crude oil crack, and soybean oil-palm oil spread arbitrage, the strategies performed well. And the project optimized the quantitative trading workflow, lowering the barrier for strategy development. The team completed the technical integration through collaboration, accumulating valuable practical experience.

Model design and Methodology

The project has adopted a modularised approach to divide the price difference arbitrage into four core modules, which are the data processing module, feature engineering module, model training module, and back-testing & evaluation module. This modularised design ensures the extensibility and maintainability of the framework, making it easier to integrate new data sources and models. A detailed breakdown of the features of the above-mentioned modules was listed below:

Data processing module

Data sources: This module is to integrate multiple data sources from various channels, which mainly includes historical futures data and open datasets related to inventory and production data.

Data cleaning: This is the core of the data processing which deals with the missing and abnormal values in the datasets. It also standardises the format of the datasets, such as aligning the time-stamps. Other functions include noise reduction and smoothing. All the standardised data was then stored in the SQLite database, in which the data could be easily retrieved by other modules.

Feature engineering module:

This is a critical part which determines the performance of the model. It focuses on extracting information with valuable predictions from the original data. Main features generated include the historical fluctuations of the price difference, which reflects the short-term market volatility; the seasonal factor, which captures the periodic change of the commodities (particularly for agricultural products); and fundamental indicators such as the inventory change rate and supply-demand ratio which drives price fluctuations.

Model training module

Gradient Boosting models like XGBoost were selected as the model for the following reasons. First, these models could deal with complex non-linear relations, which is common in the price-difference scenario. Secondly, they have shown robustness with missing and abnormal values in the dataset. To improve the generalisation capacity of the model, hyperparameter optimisation like grid search and cross-validation was combined to automatically refine the hyperparameters of the model.

Backtesting & Evaluation module

This module is to validate the strategy. It supports calculations of key performance metrics including the annualised returns which measures long-term profitability, sharpe ratio, which evaluates the return after risk adjustment, and the maximum drawdown which reflects the possible greatest risk-level. To simulate a real world trading scenario while mitigating risks, two risk-control mechanisms were included in the framework, namely stop-loss threshold to automatically close the position when a single trade loss reaches the preset ratio, and position management, which dynamically adjusts the position based on market volatility.

Experiments and Results Analysis

This chapter presents a systematic analysis of data, features, model performance, and backtesting results based on three main experimental cases in the project: coking coal-coke spread, crude oil crack spread, and soybean-palm oil spread. Each subsection provides methodology descriptions, evaluation metrics, and discussion points. For detailed numerical data, charts, and complete backtesting reports, please refer to the corresponding notebook files (`trainmodel_coking_spread.ipynb` , `trainmodel_crack_spread.ipynb` , `trainmodel_sp_spread.ipynb`) and the `*_Results/strategy_runs/` subdirectories under the `outputs/` directory.

Case Study 1: Coking Coal-Coke Spread Arbitrage (`trainmodel_coking_spread.ipynb`)

Data and Features

The primary data used in this case includes futures prices (main continuous contracts), exchange warehouse receipt data, and publicly available fundamental data. Data preprocessing involves aligning trading calendars, filling missing values, handling contract rollovers (rolling/adjust), and unit standardization. Key features include:

- Lagged terms and moving averages of the spread series (short-term and long-term windows).
- Volatility features (e.g., rolling n-day standard deviation).
- Fundamental-derived indicators (inventory change rate, warehouse receipt ratio).
- Statistical indicators of individual commodity prices.

Model Training and Feature Importance

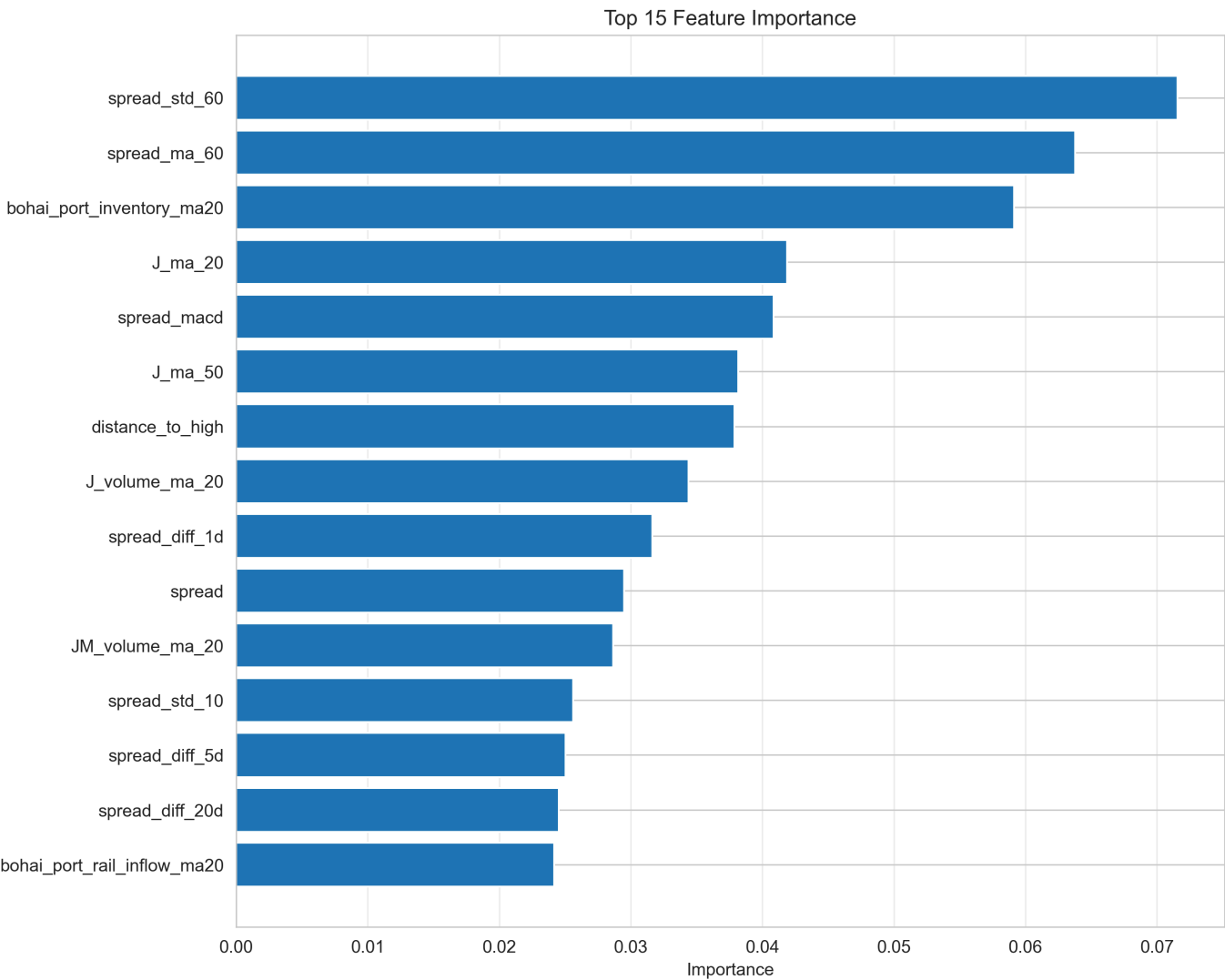


Figure 1: Feature Importance Analysis for Coking Coal-Coke Spread Strategy

Figure 1 displays the relative importance ranking of features in the coking coal-coke spread prediction model. As shown in the figure, historical statistical features of the spread (such as rolling mean and standard deviation) and fundamental indicators (such as inventory change rate) contribute significantly to model predictions, validating the core role of mean reversion characteristics and fundamental factors in spread arbitrage strategies.

Backtesting Results Analysis

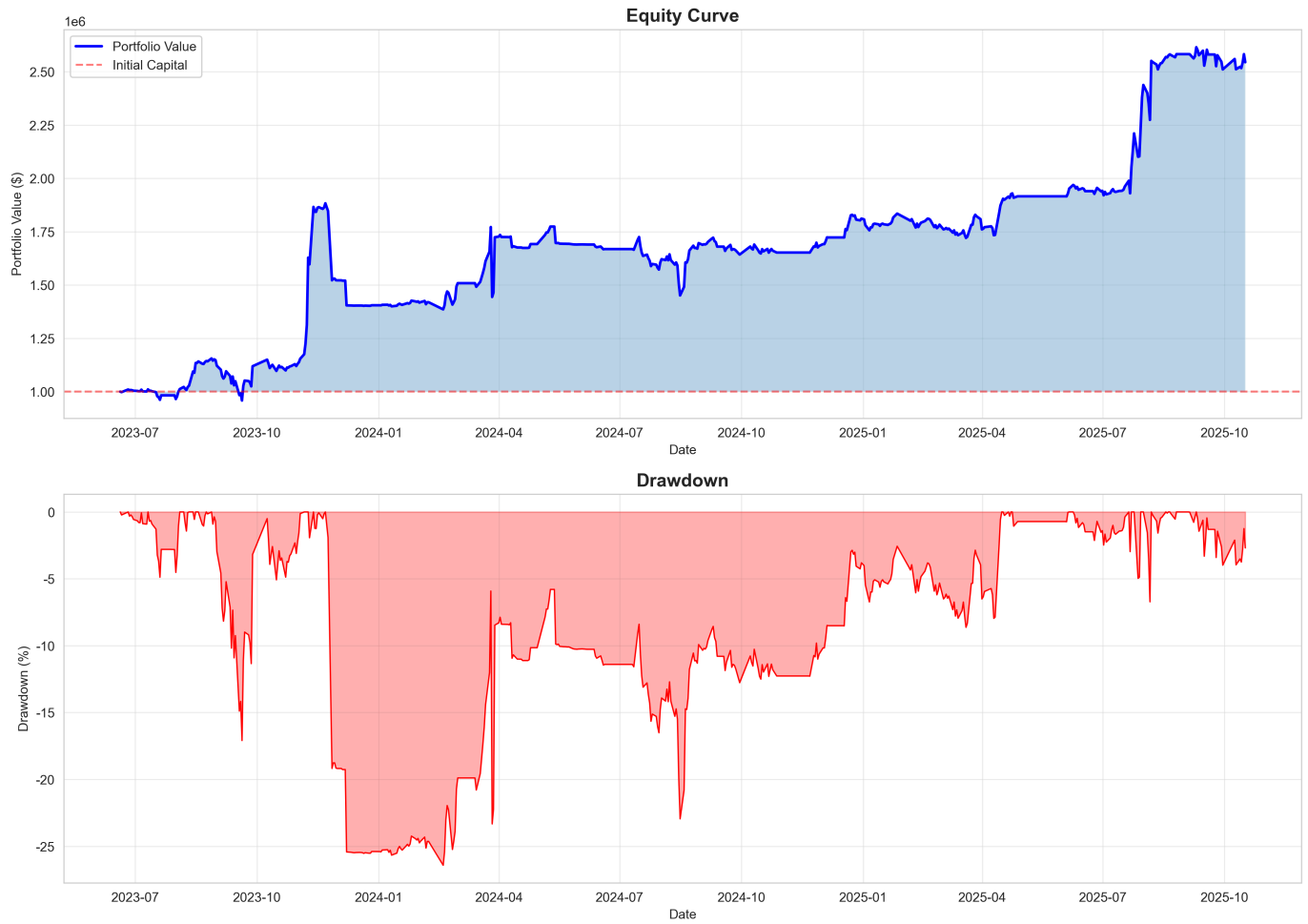


Figure 2: Equity Curve for Coking Coal-Coke Spread Strategy

Figure 2 presents the equity curve and drawdown trajectory of the coking coal-coke arbitrage strategy. The equity curve shows a steady upward trend overall, with a cumulative return of 154.49% and an annualized return of 49.39%. The drawdown curve indicates a maximum drawdown of -26.42%, occurring during periods of severe market volatility, but overall risk control performance is strong, demonstrating the strategy's robust risk-adjusted return capability.

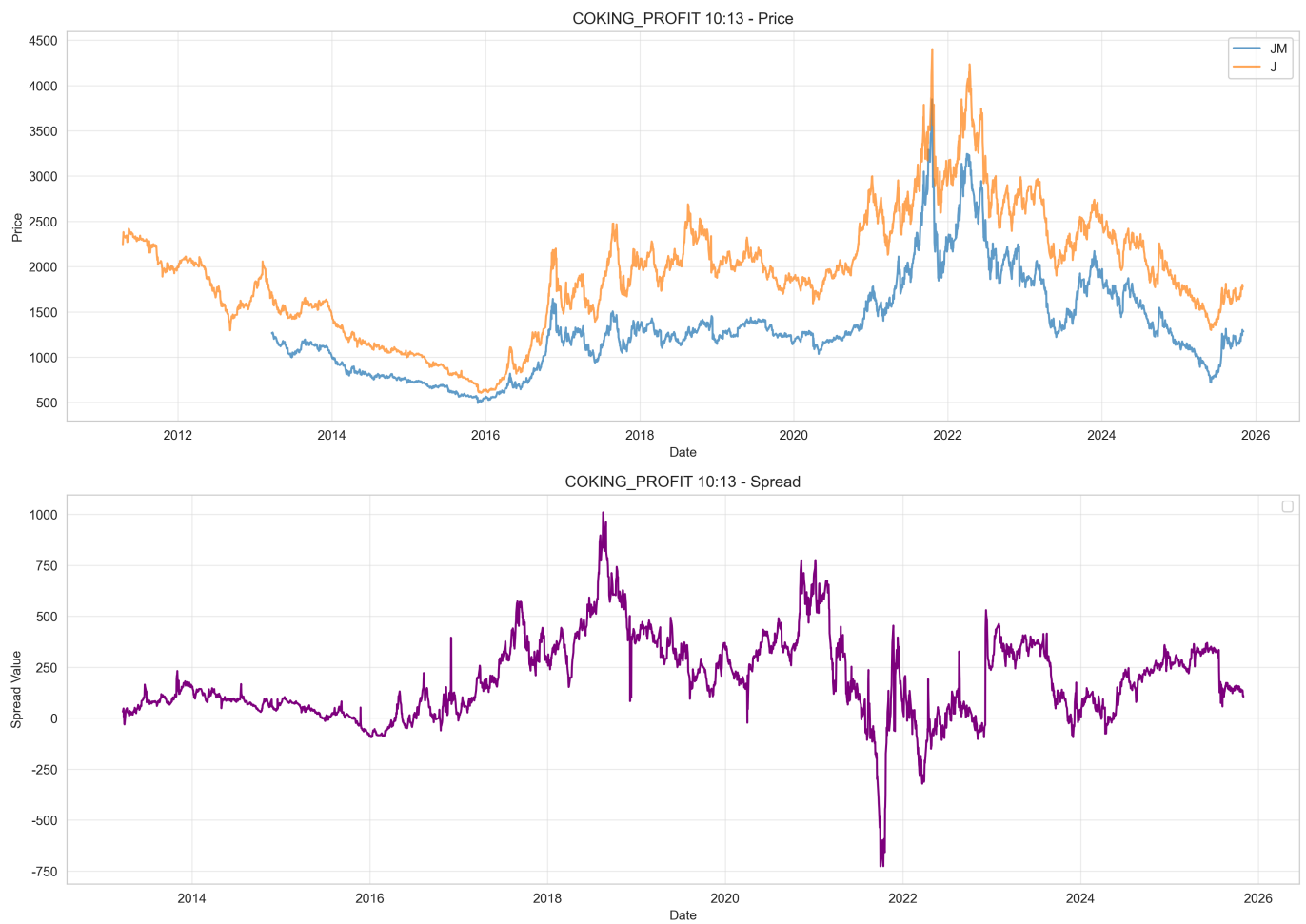


Figure 3: Price and Spread Trends for Coking Coal-Coke

Figure 3 illustrates the price trends of coking coal and coke, as well as their spread dynamics. The upper section displays the price time series of both commodities, while the lower section shows the dynamic evolution of the spread and trading signal trigger points. The figure reveals the spread's fluctuation characteristics around its mean, as well as the strategy's entry and exit timing when the spread deviates from threshold levels.

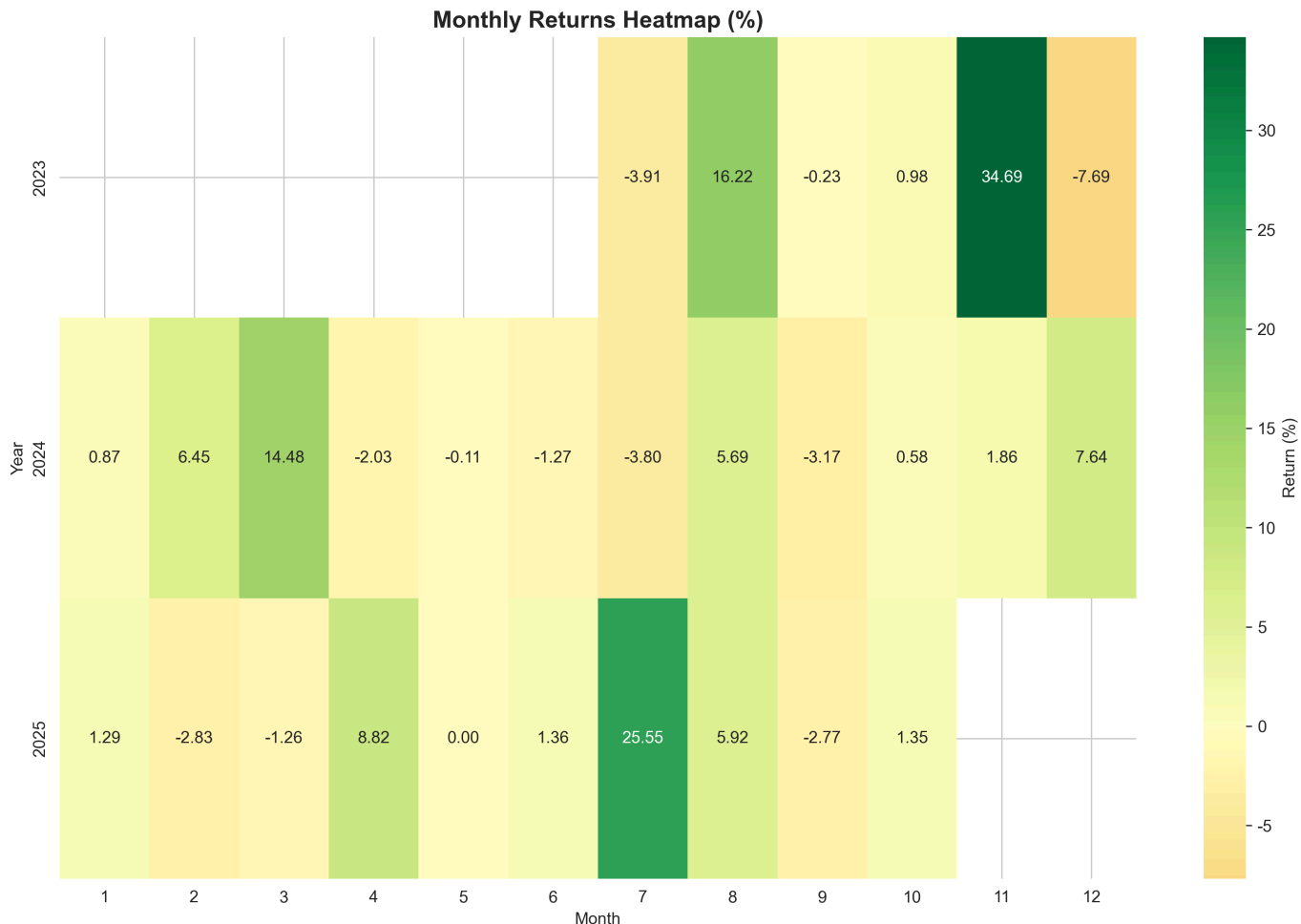


Figure 4: Monthly Returns Heatmap for Coking Coal-Coke Strategy

Figure 4 presents the strategy's return performance across different years and months in heatmap format. Color intensity represents return magnitude, enabling identification of seasonal characteristics and stability under different market conditions. Overall, the strategy performs stably in most months, with larger losses concentrated in periods of severe market volatility.

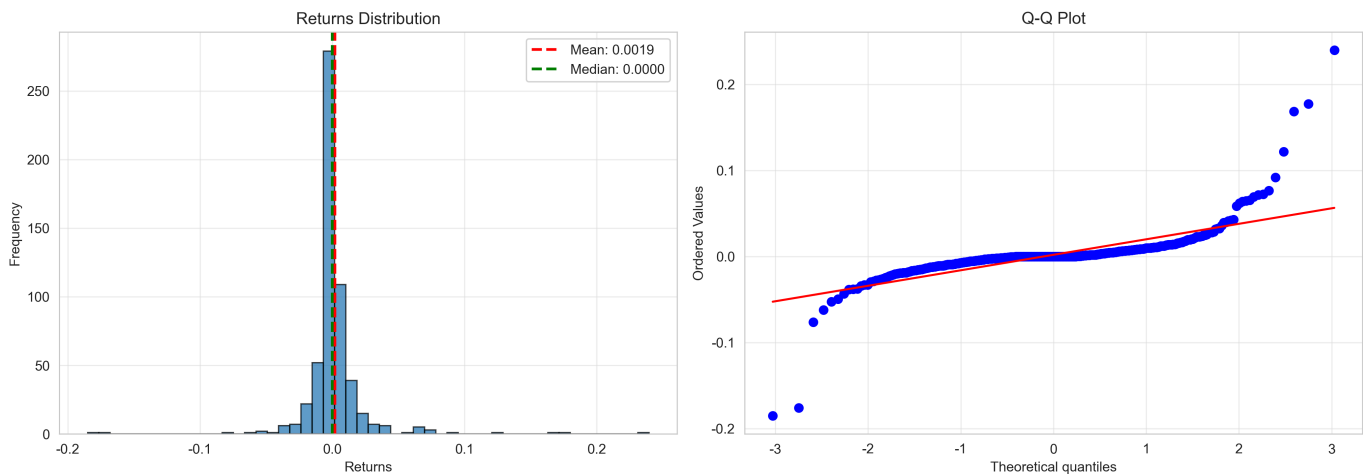


Figure 5: Returns Distribution for Coking Coal-Coke Strategy

Figure 5 displays the distribution characteristics of the strategy's daily returns. The left panel shows a histogram with kernel density estimation, while the right panel presents a Q-Q plot to test return normality. The distribution exhibits a slight positive skew with relatively limited tail risk, consistent with typical characteristics of spread arbitrage strategies.

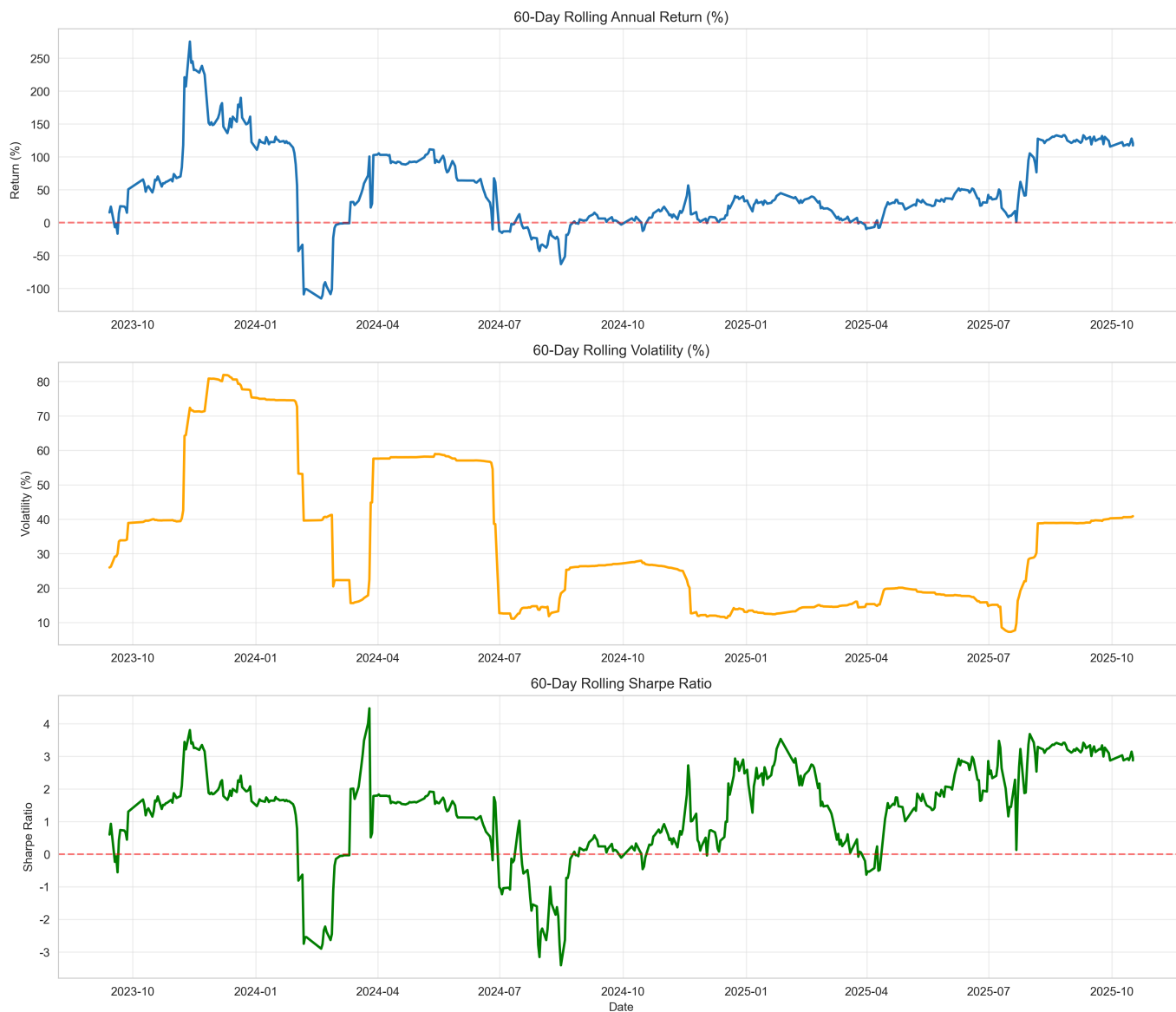


Figure 6: Rolling Performance Metrics for Coking Coal-Coke Strategy

Figure 6 shows the evolution of rolling window performance metrics for the strategy, including rolling Sharpe ratio, rolling maximum drawdown, and other key indicators. By observing the temporal evolution of these metrics, one can assess the strategy's stability and adaptability across different market cycles. The figure demonstrates that the Sharpe ratio maintains a relatively high level in most periods, indicating sustained risk-adjusted returns.

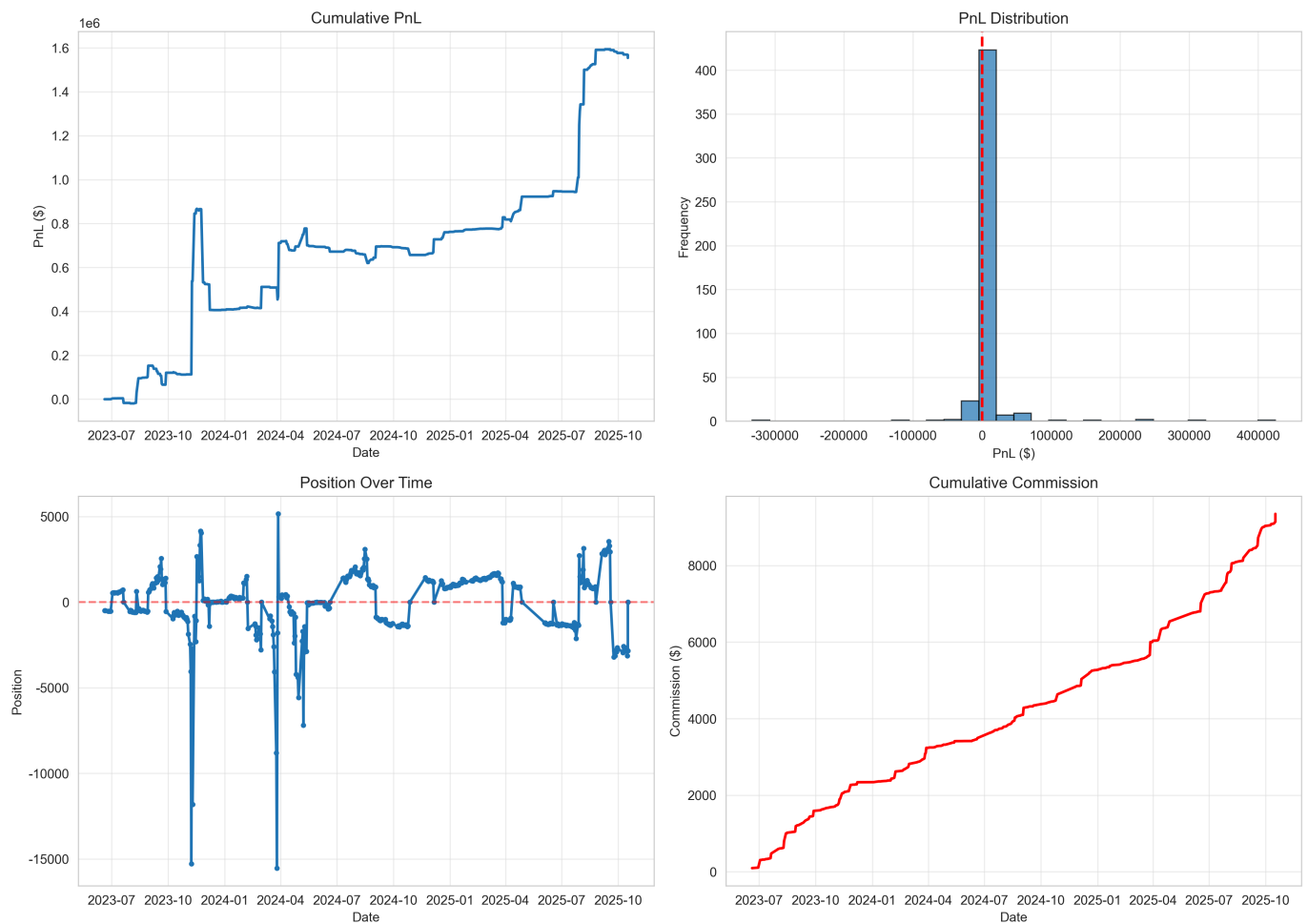


Figure 7: Trade Analysis for Coking Coal-Coke Strategy

Figure 7 provides a detailed analysis of the strategy's trading behavior, including trade frequency distribution, holding period distribution, and profit/loss distribution per trade. The figure reveals that the strategy achieves a win rate of 55.76% and a profit-loss ratio of 2.60, indicating that the strategy generates positive expected returns by controlling individual losses and amplifying individual profits. The holding period distribution shows the strategy primarily maintains short-to-medium-term positions, consistent with the time scale of spread mean reversion.

Case Study 2: Crude Oil Crack Spread (`trainmodel_crack_spread.ipynb`)

Data and Features

The crude oil crack spread involves calculating the spread between crude oil and refined products (such as diesel and gasoline), with data including main contract prices and macroeconomic fundamental data. Preprocessing also covers contract rollovers and unit conversions (e.g., converting crude oil barrels and refined products by energy or volume). Features focus on:

- Historical statistics of crack spread (mean, skewness, kurtosis).
- Daily macroeconomic factors such as the US Dollar Index and VIX.
- Statistical indicators of individual commodity prices.
- Conversions across different pricing units.

Model Training and Feature Importance

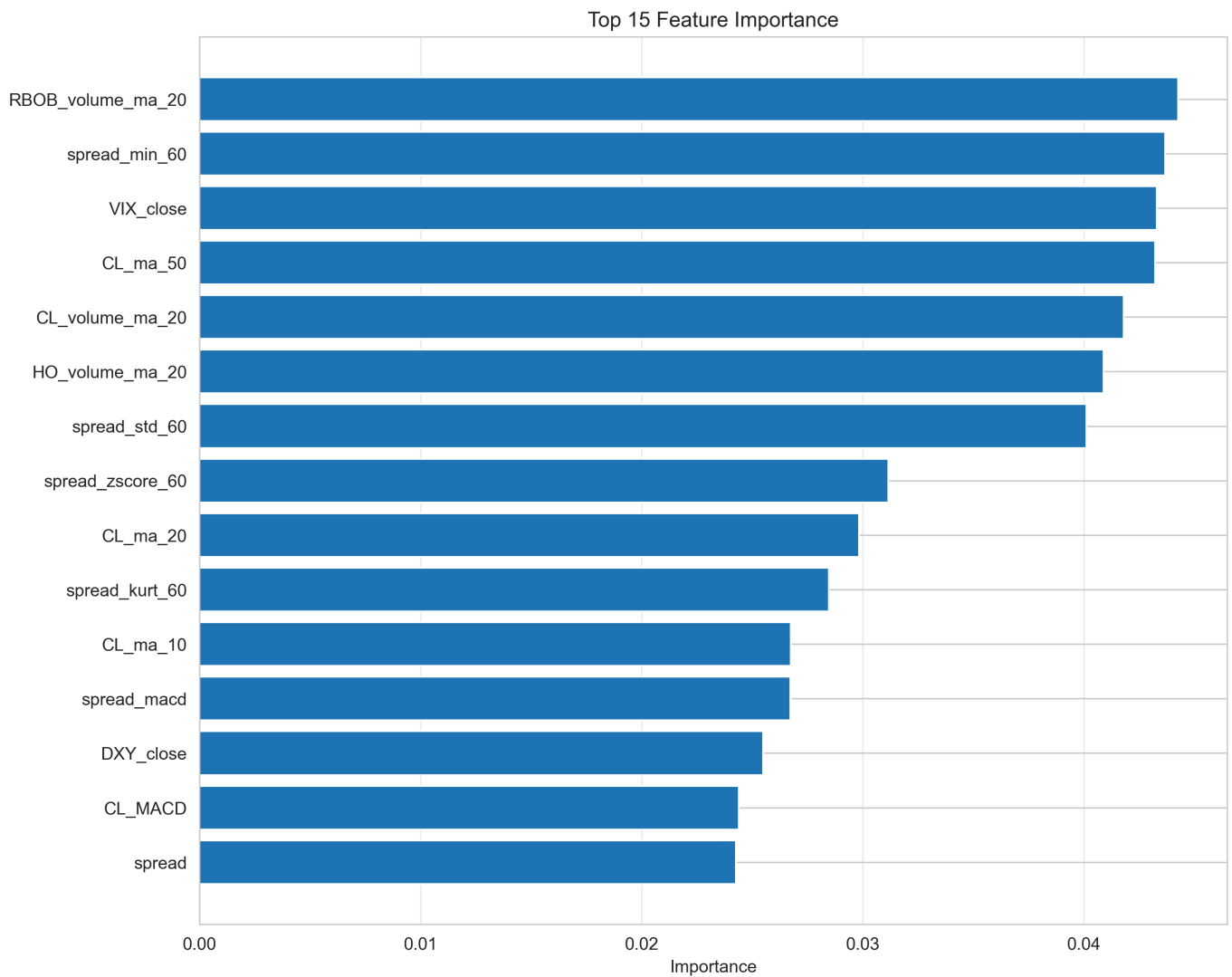


Figure 8: Feature Importance Analysis for Crack Spread Strategy

Figure 8 displays the feature importance ranking for the crack spread prediction model. Compared to the coking coal-coke case, macroeconomic factors (such as VIX) are more prominent in the crack spread model, reflecting the tight connection between energy markets and the global macroeconomy. Historical volatility of the spread also contributes significantly to predictions.

Backtesting Results Analysis

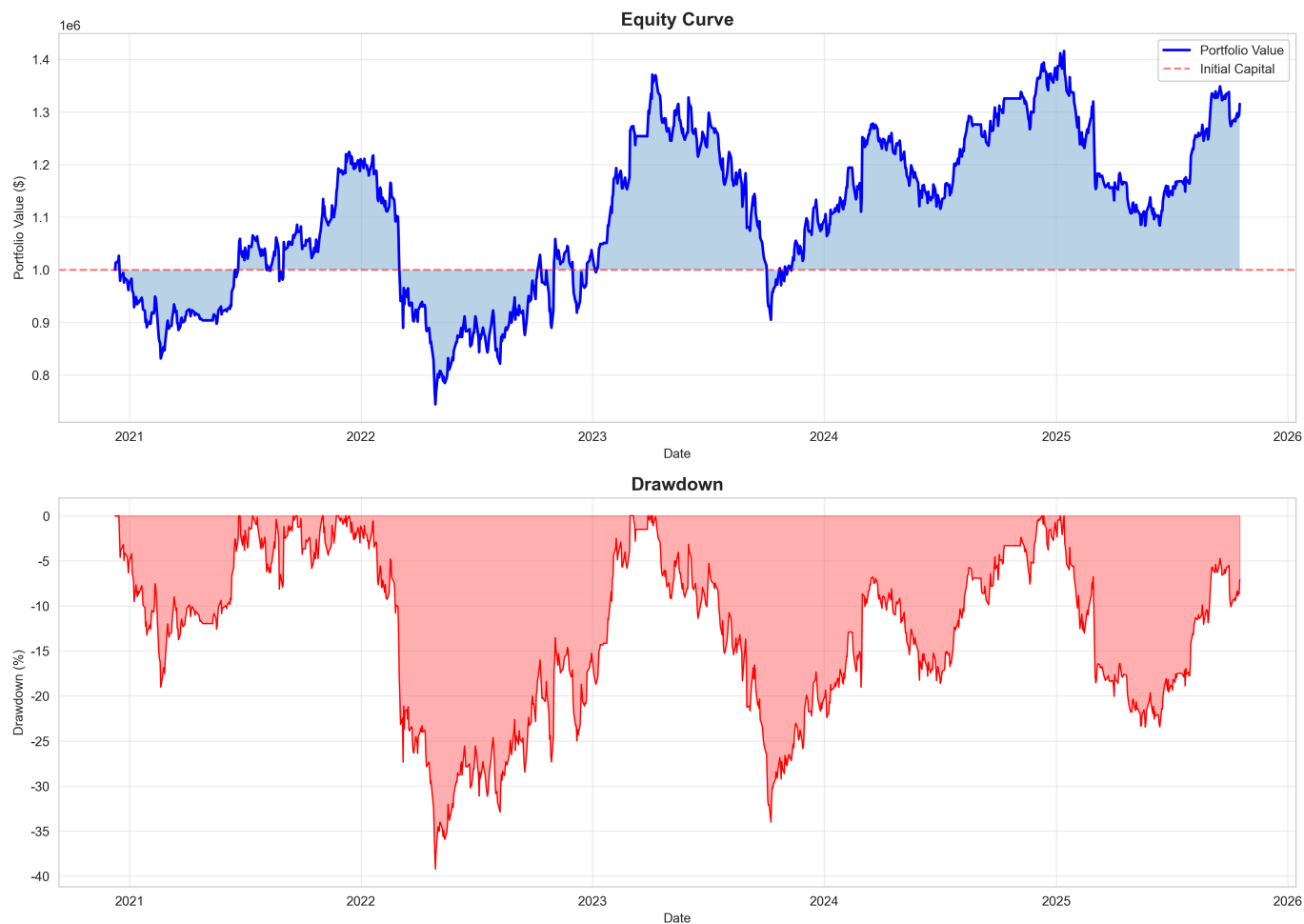


Figure 9: Equity Curve for Crack Spread Strategy

Figure 9 presents the equity curve and drawdown trajectory of the crack spread arbitrage strategy. Compared to the coking coal-coke strategy, this strategy exhibits greater equity curve volatility, with a cumulative return of 31.50% and an annualized return of only 5.81%, showing relatively modest performance.

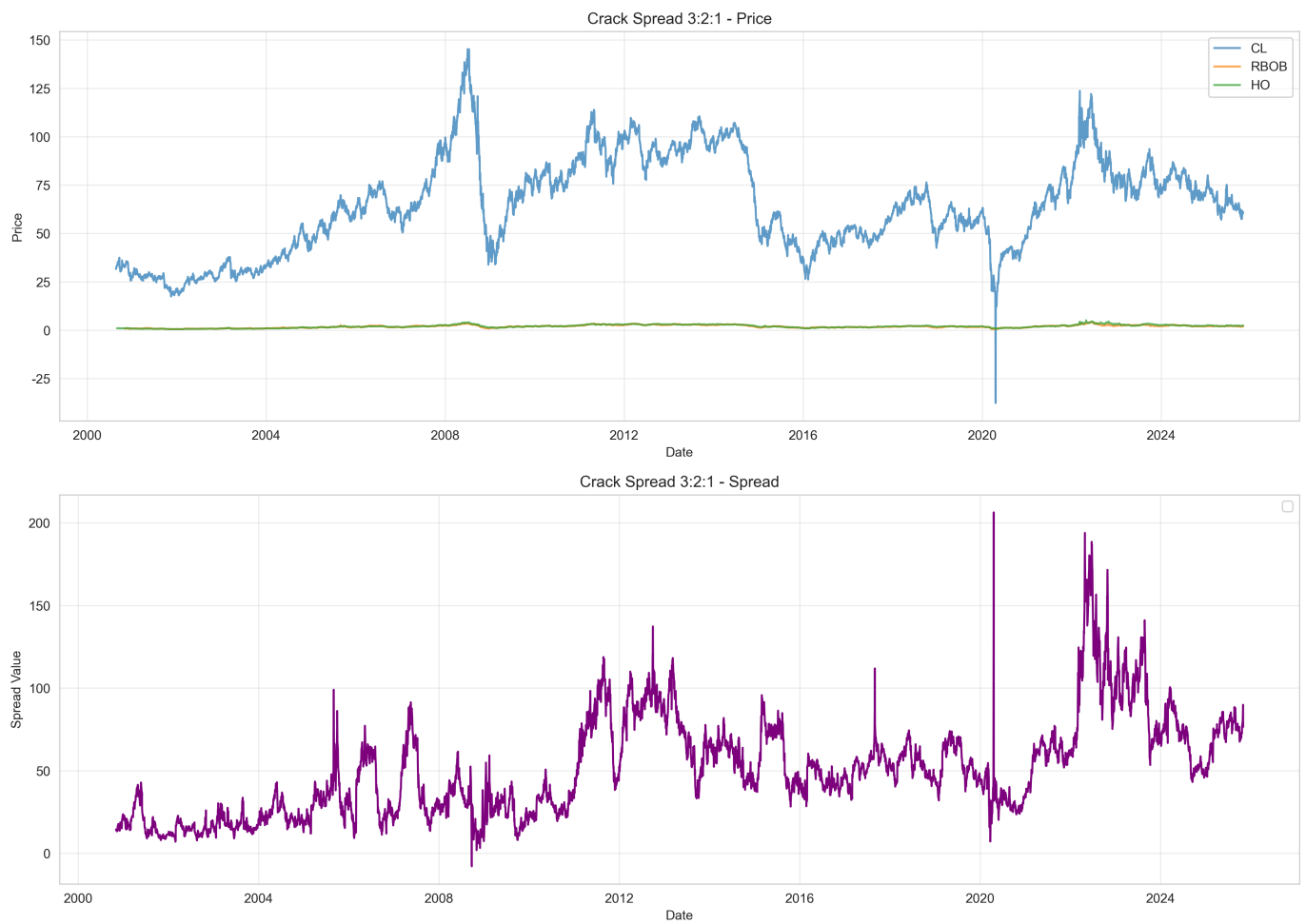


Figure 10: Price and Spread Trends for Crack Spread

Figure 10 illustrates the price trends of crude oil and refined products, as well as the dynamic evolution of the crack spread. The figure reveals the severe volatility characteristics unique to energy markets, particularly during geopolitical events and pandemic shocks, when the spread exhibited extreme deviations beyond historical ranges. This non-stationary characteristic increases the risk of mean reversion strategies and explains the strategy's subpar performance.

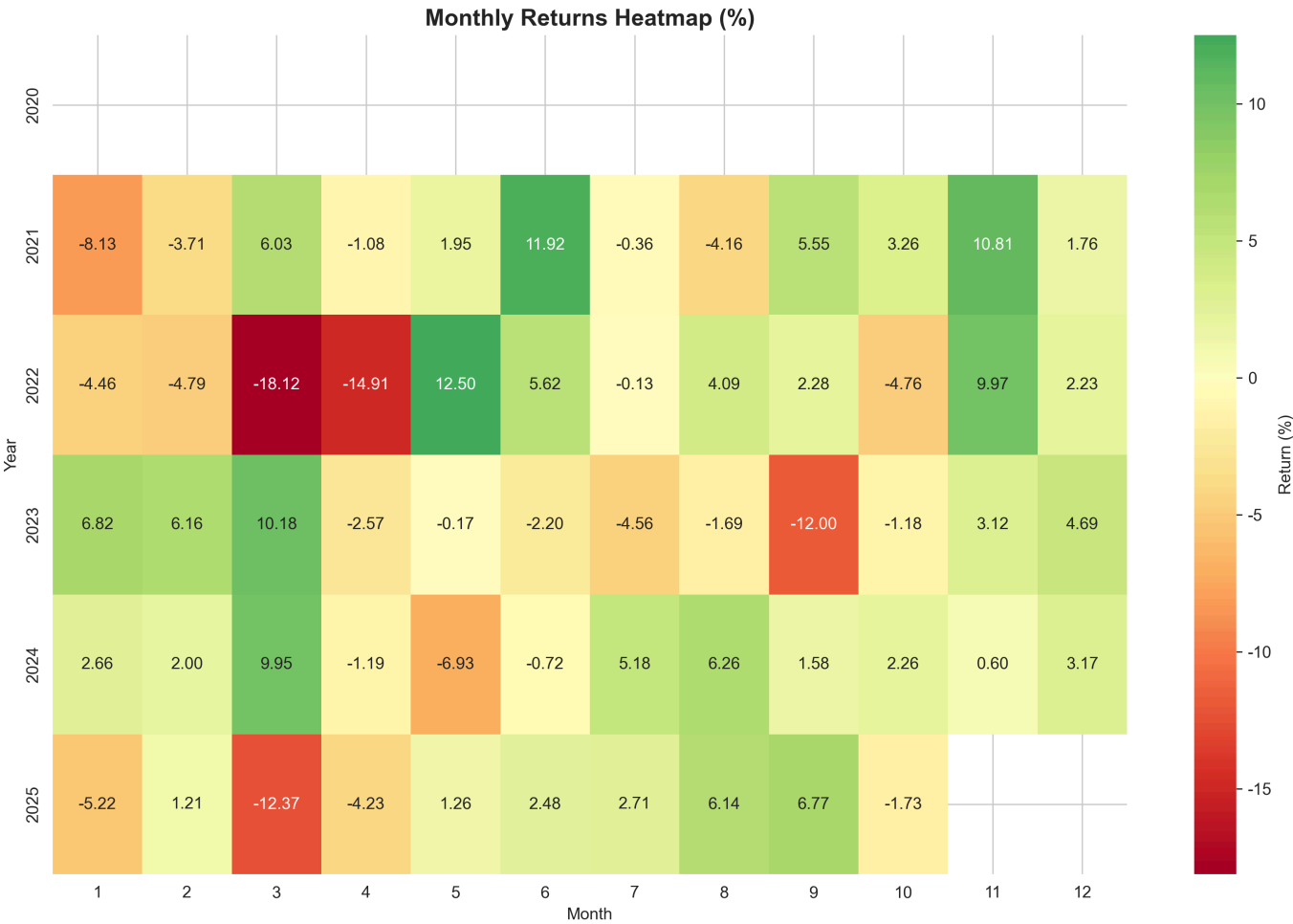


Figure 11: Monthly Returns Heatmap for Crack Spread Strategy

Figure 11 presents the monthly return distribution of the crack spread strategy in heatmap format. The figure clearly shows significant losses (deep red areas) in March-April 2020, corresponding to the crude oil price collapse and negative oil price event caused by the pandemic. Apart from this, the strategy performs steadily in most months but lacks periods of concentrated positive returns, resulting in overall low returns.

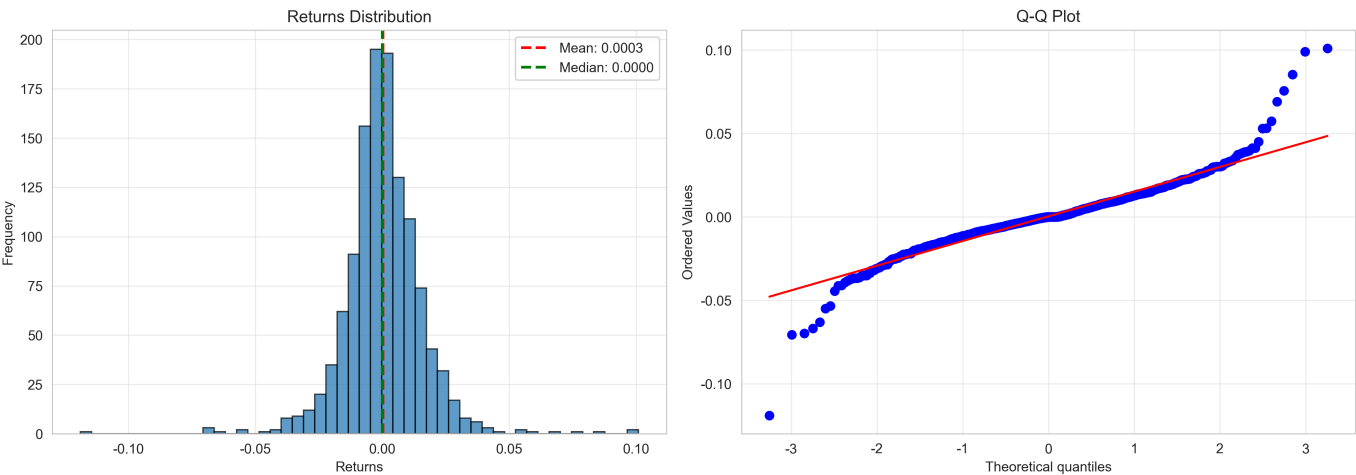


Figure 12: Returns Distribution for Crack Spread Strategy

Figure 12 displays the distribution characteristics of the crack spread strategy's daily returns. Compared to the coking coal-coke strategy, this strategy's return distribution exhibits more pronounced negative skewness and fat tail characteristics, indicating higher tail risk exposure. The Q-Q plot shows clear deviations from normal distribution, with extreme loss events occurring more frequently than normal assumptions would suggest, posing higher requirements for risk management.

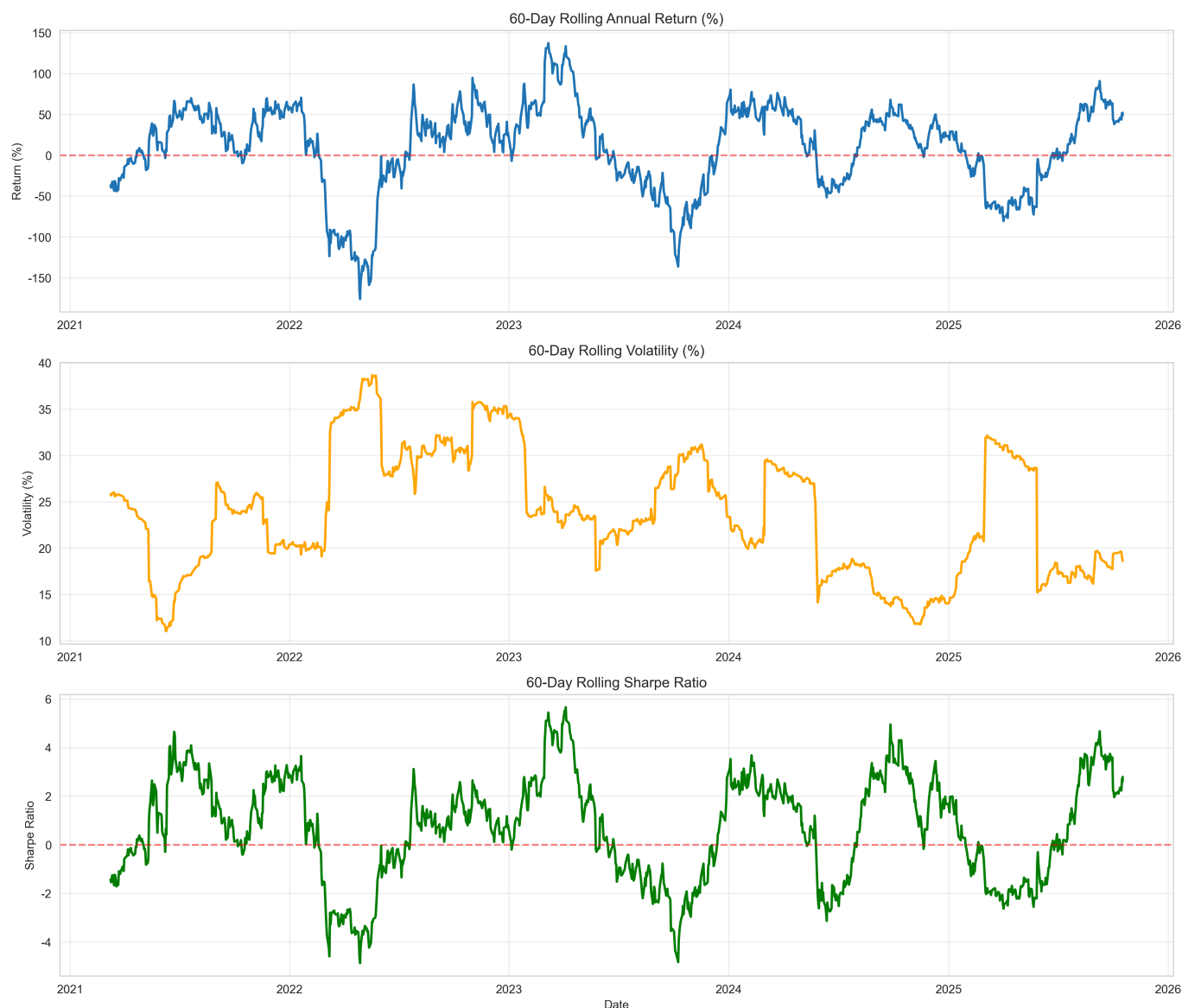


Figure 13: Rolling Performance Metrics for Crack Spread Strategy

Figure 13 shows the evolution of rolling window performance metrics for the strategy. The figure indicates that the Sharpe ratio fluctuates significantly across different periods, performing adequately during stable market conditions but deteriorating sharply during high volatility periods. Rolling maximum drawdown metrics also reveal the strategy's lack of effective risk control mechanisms under extreme market conditions, which is a key area requiring improvement. To address this, we will introduce a large language model-based risk identification mechanism in later sections to assist in trade signal execution, thereby mitigating drawdown risks arising from historically unprecedented extreme events.

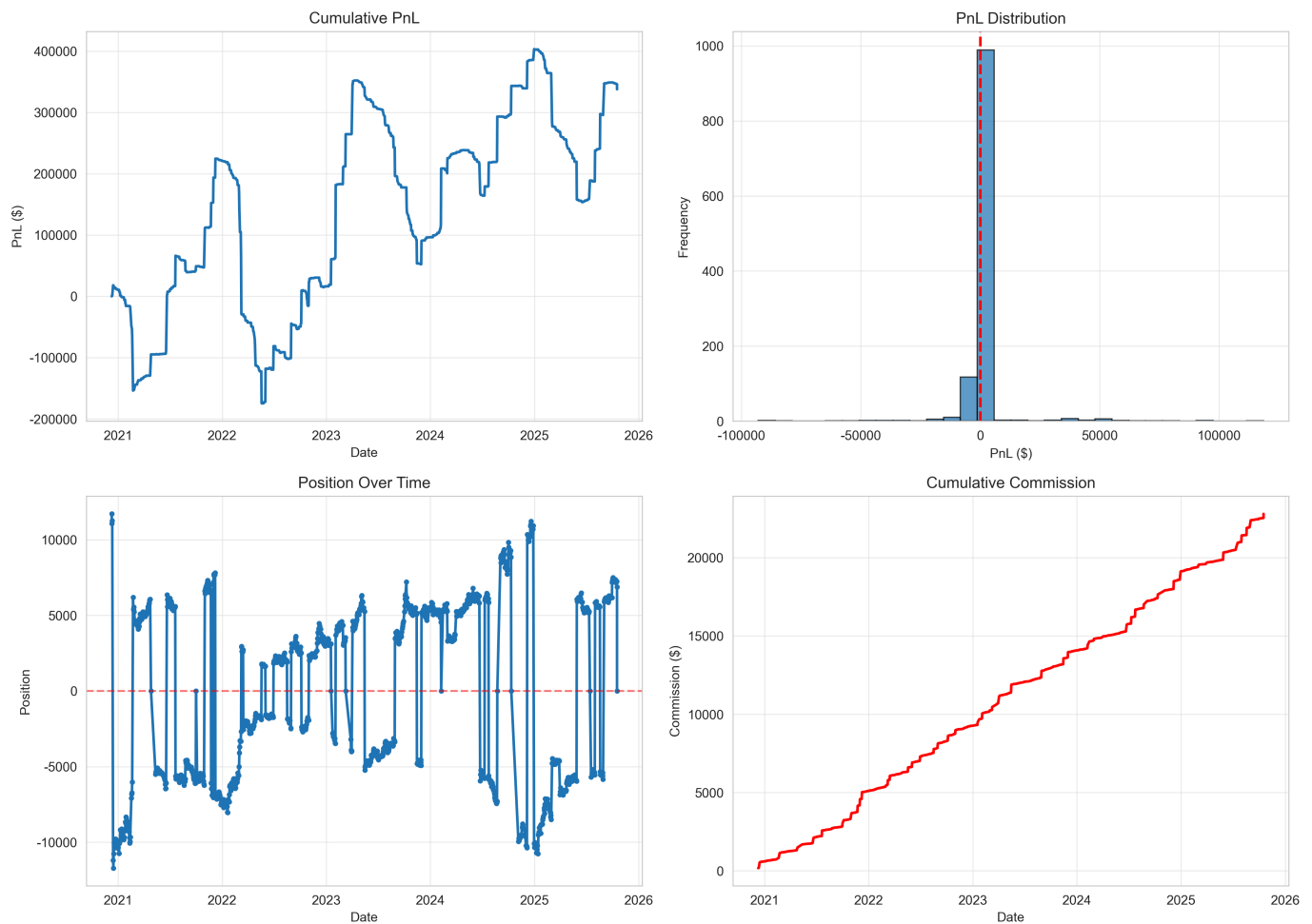


Figure 14: Trade Analysis for Crack Spread Strategy

Figure 14 provides a detailed analysis of the strategy's trading behavior. The strategy achieves a win rate of only 38.35%, significantly lower than the other two cases, with a profit-loss ratio of 1.25, indicating the strategy's failure to effectively control individual losses or amplify individual profits. With 1,167 trades, the highest among the three strategies, transaction costs (commission fees of 22,799 yuan) increase substantially. The combination of excessive trading frequency and low win rate is the primary reason for this strategy's poor performance.

Case Study 3: Soybean-Palm Oil Spread (`trainmodel_sp_spread.ipynb`)

Data and Features

The challenges of cross-commodity arbitrage primarily stem from differences in trading hours, trading regulations, and fundamental structures across markets. Data includes main contract futures prices, port inventory, and import-export data. Key features include:

- Lagged statistical features of cross-market adjusted spreads.
- Daily macroeconomic factors such as the US Dollar Index and VIX.
- Fundamental data such as inventory levels.

Model Training and Feature Importance

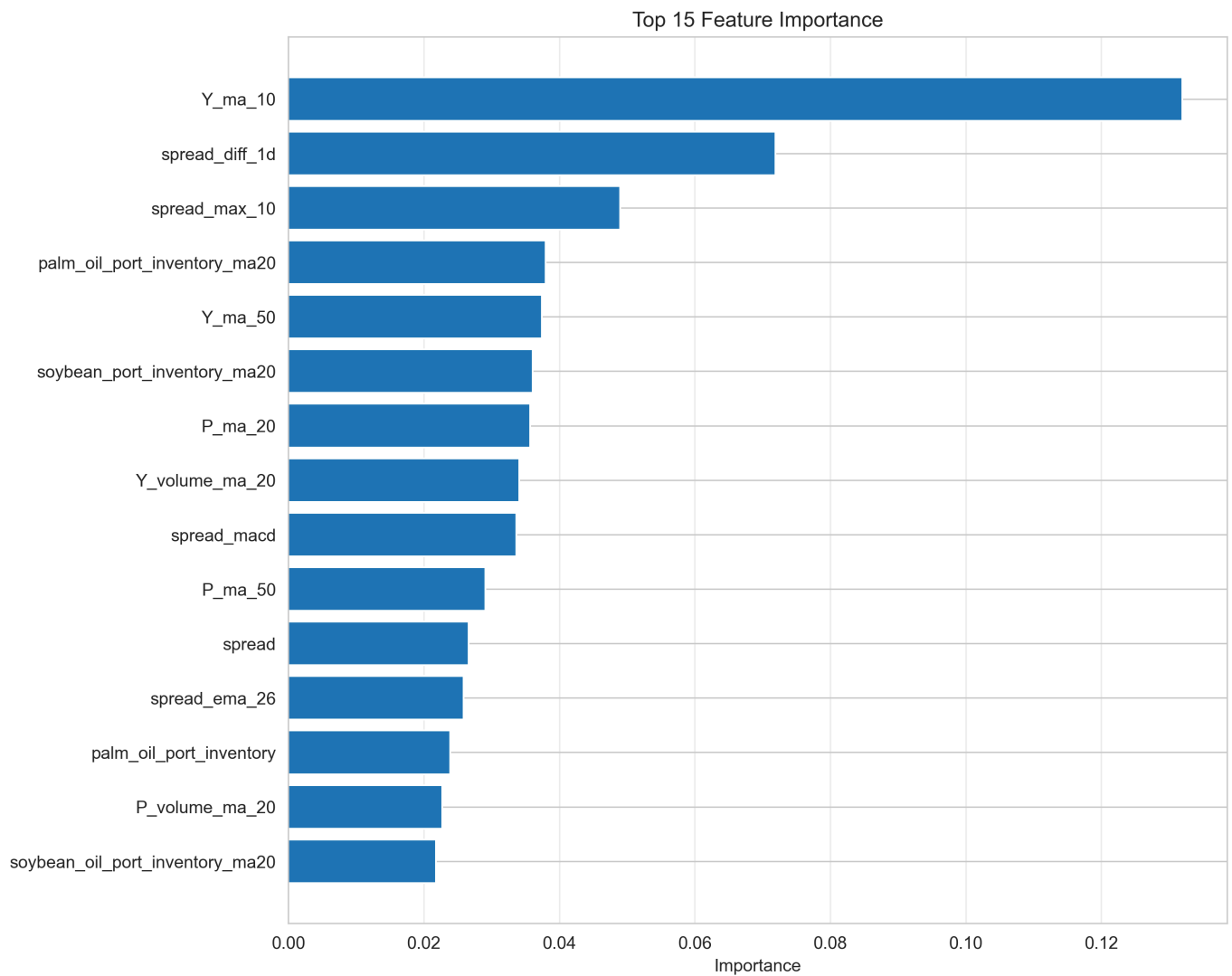


Figure 15: Feature Importance Analysis for Soybean-Palm Oil Spread Strategy

Figure 15 displays the feature importance ranking for the soybean-palm oil spread prediction model. Due to the cross-commodity arbitrage nature, price momentum features and relative strength indicators for both commodities occupy important positions in the model. Additionally, port inventory data contributes significantly to spread predictions, reflecting the important influence of supply-demand cycles in agricultural commodity markets.

Backtesting Results Analysis

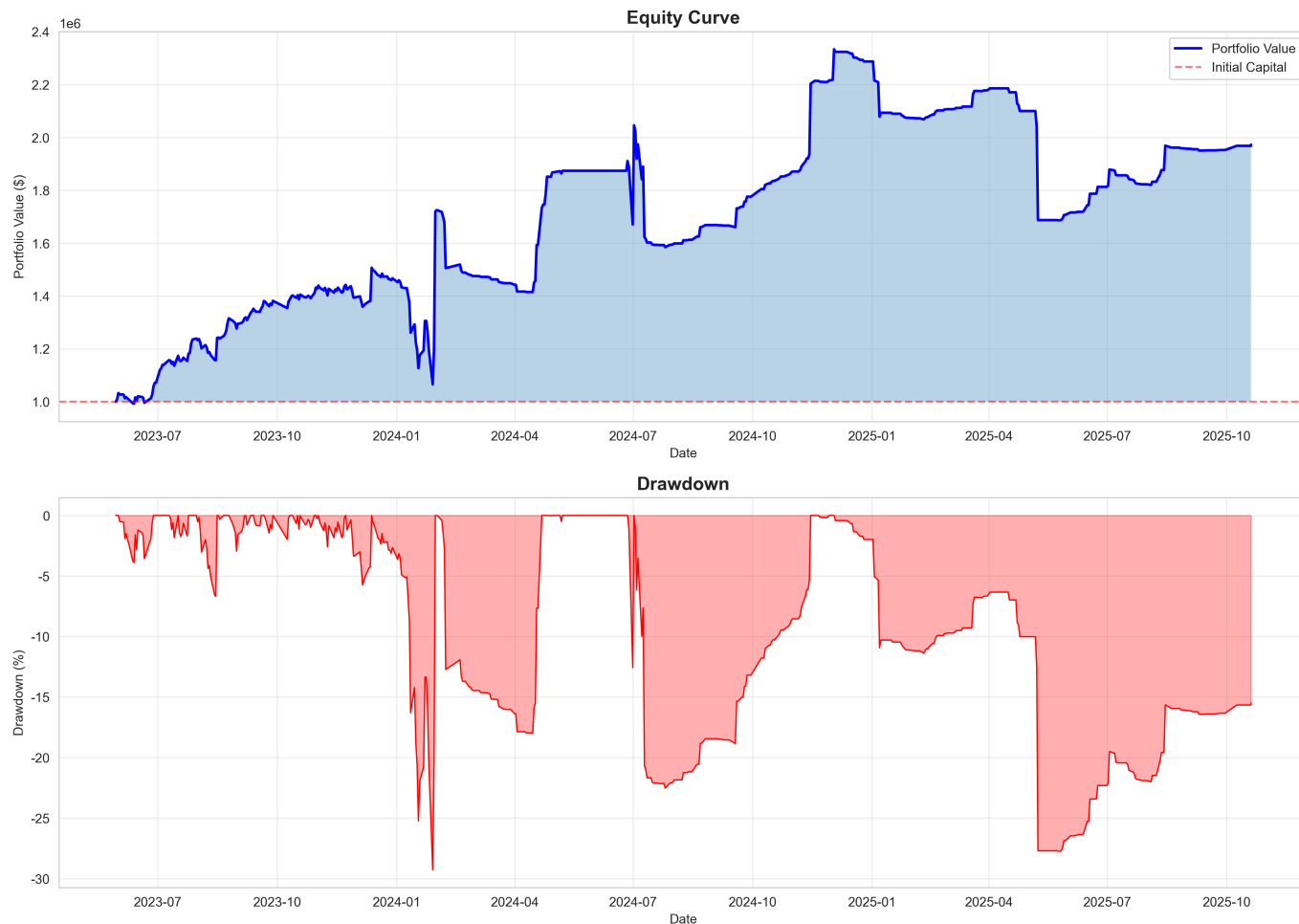


Figure 16: Equity Curve for Soybean-Palm Oil Spread Strategy

Figure 16 presents the equity curve and drawdown trajectory of the soybean-palm oil arbitrage strategy. The equity curve exhibits an overall upward trend, with a cumulative return of 97.28% and an annualized return of 32.84%, performing between the coking coal-coke and crack spread strategies. The drawdown curve shows a maximum drawdown of -29.27%, indicating moderate risk levels. Notably, the strategy performed exceptionally well after 2020, possibly related to changes in global vegetable oil supply-demand dynamics following the pandemic.

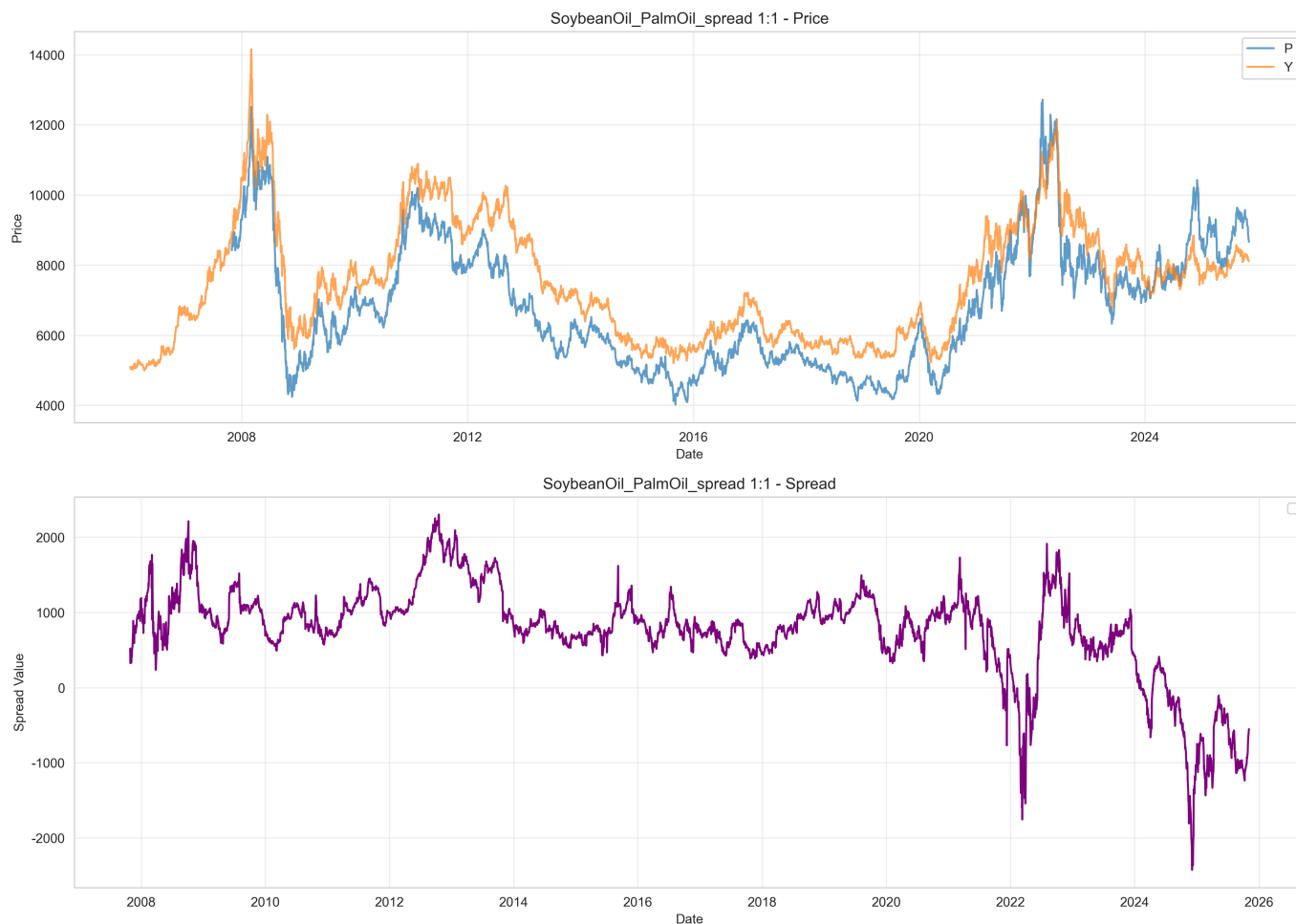


Figure 17: Price and Spread Trends for Soybean-Palm Oil

Figure 17 illustrates the price trends of soybean oil and palm oil, as well as the dynamic evolution of their spread. The figure reveals strong co-movement in the price trends of both commodities, but during specific periods (such as palm oil production declines due to abnormal weather in Southeast Asia, or soybean import impacts from US-China trade tensions), significant spread widening or narrowing occurs. The strategy generates returns by capturing these spread deviation and reversion opportunities.



Figure 18: Monthly Returns Heatmap for Soybean-Palm Oil Strategy

Figure 18 presents the strategy's monthly return distribution in heatmap format. The strategy also experiences some loss months (red areas), but profitable months predominate overall, demonstrating good stability.

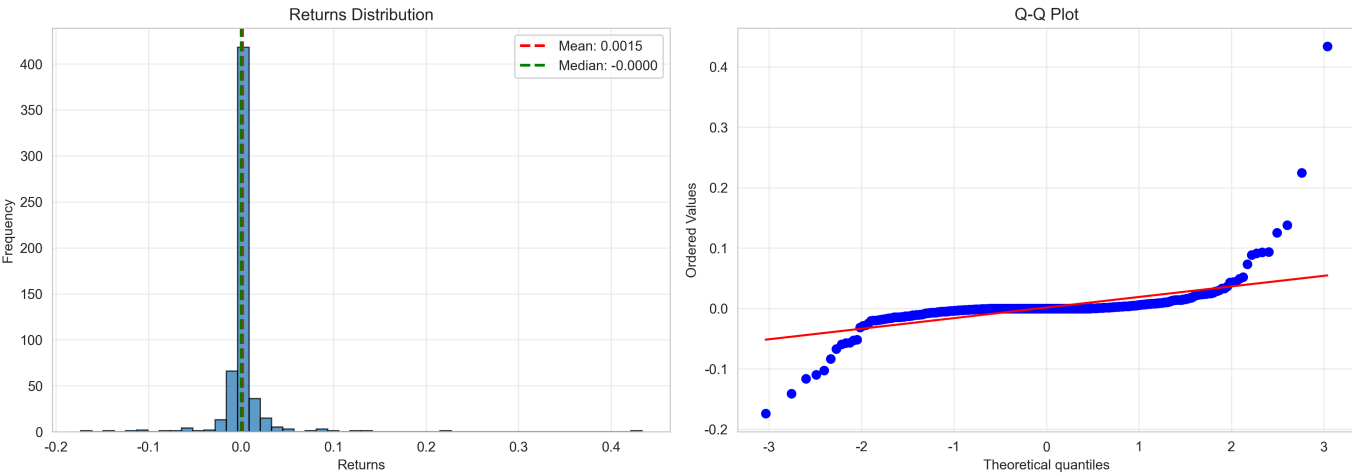


Figure 19: Returns Distribution for Soybean-Palm Oil Strategy

Figure 19 displays the distribution characteristics of the strategy's daily returns. Similar to the coking coal-coke strategy, the return distribution approximates normality but exhibits some fat tail characteristics. A slightly thicker left tail indicates the strategy's risk exposure during extreme downturns, consistent with the soybean-palm oil spread's higher annualized volatility (45.18%). The Q-Q plot shows deviations from normal distribution in extreme value regions, but overall fit is superior to the crack spread strategy.

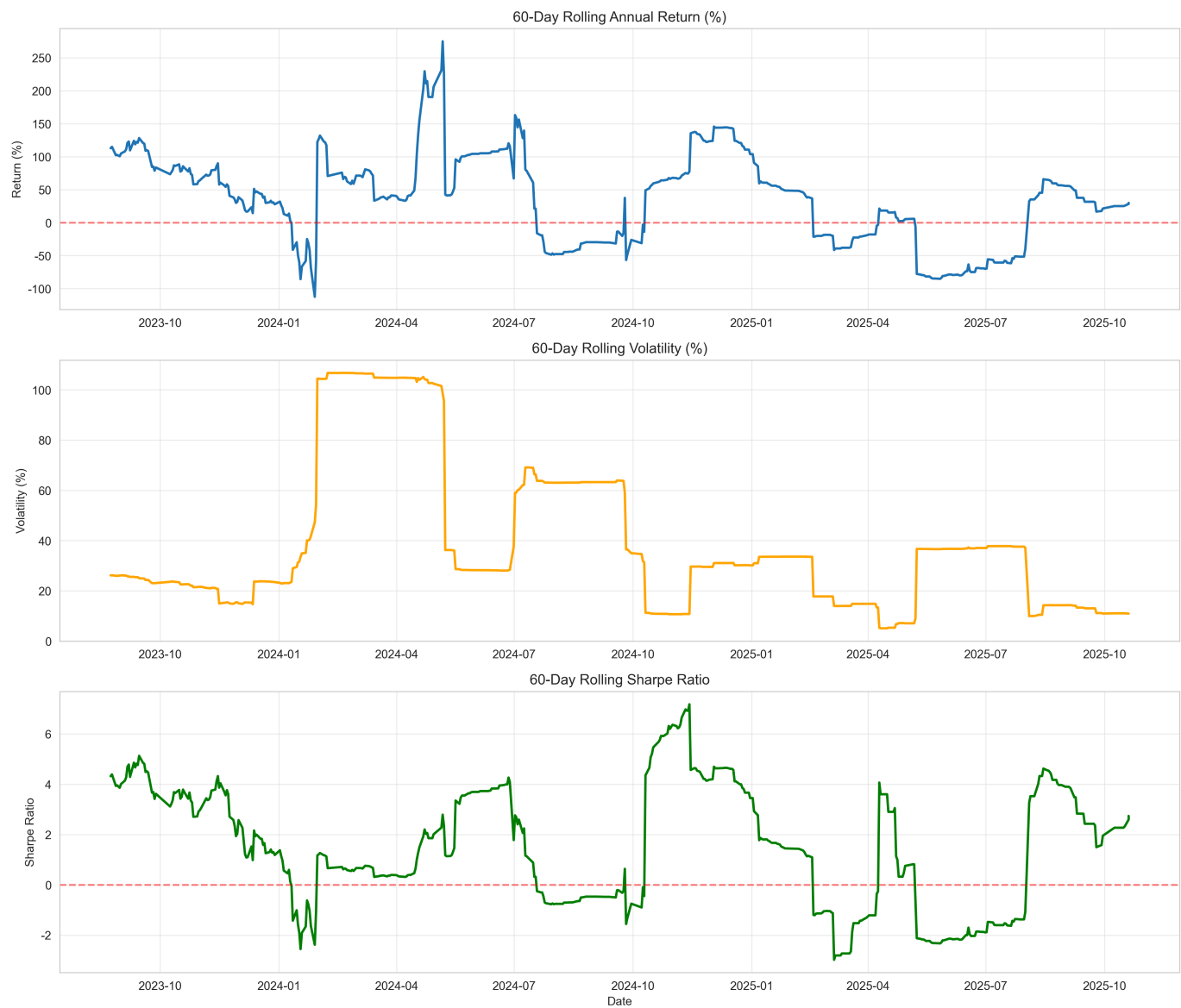


Figure 20: Rolling Performance Metrics for Soybean-Palm Oil Strategy

Figure 20 shows the evolution of rolling window performance metrics for the strategy. The Sharpe ratio exhibits an upward trend during the backtesting period, maintaining particularly high levels after 2020, reflecting the strategy's good adaptability to changing market conditions. Rolling maximum drawdown metrics show the strategy experienced multiple moderate-scale drawdowns, but recovered within reasonable timeframes in each case, demonstrating the effectiveness of spread mean reversion characteristics.

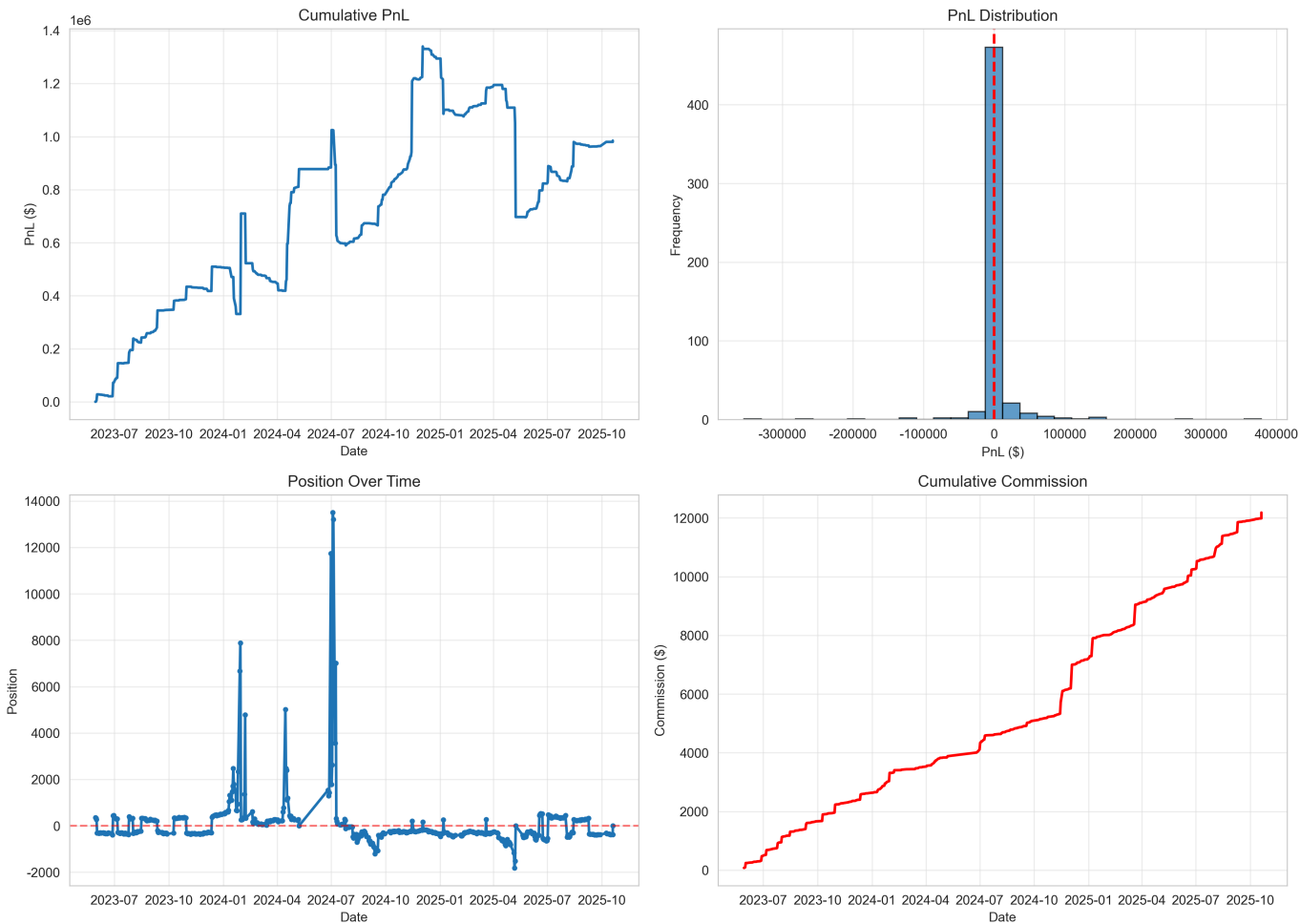


Figure 21: Trade Analysis for Soybean-Palm Oil Strategy

Figure 21 provides a detailed analysis of the strategy's trading behavior. The strategy achieves a win rate of 54.92% and a profit-loss ratio of 1.54, demonstrating balanced and reasonable performance. With 533 trades, the frequency is at a moderate level, ensuring the strategy captures sufficient trading opportunities while avoiding cost erosion from overtrading. The holding period distribution shows the strategy primarily maintains medium-term positions, matching the adjustment cycles of agricultural commodity spreads.

Comprehensive Comparison

Cross-case comparison among the three cases helps understand the generalizability and limitations of models and strategies. Comparison dimensions include: annualized returns, Sharpe ratio, maximum drawdown, win rate, strategy stability (volatility across different backtesting periods), and resilience to extreme events.

Model Performance

Training employs time series cross-validation (TimeSeriesSplit) combined with gradient boosting methods (XGBoost), targeting the prediction of spread risk-reward ratios (interval mean/standard deviation) over future periods. Evaluation metrics include: MSE, MAE, etc. Of course. Here is the information presented in English, with a concise explanation in paragraph form.

Model Evaluation Results (Out-of-Sample)

Model Name	MSE	RMSE	MAE	R ²
Coking Coal-Coke Spread	5.491222	2.343336	1.878188	-0.2415
Cracking Spread	9.105816	3.017585	2.458199	-0.7070
Soybean-Palm Spread	7.588987	2.754812	2.208384	-0.2108

Analysis and Interpretation

The evaluation metrics indicate a poor fit for all three XGBoost models on the out-of-sample data. The negative R² values signify that the models' predictions are worse than simply using the mean of the actual values for forecasting. High MSE, RMSE, and MAE values further confirm significant prediction errors.

However, in the context of statistical arbitrage, traditional goodness-of-fit metrics like these do not tell the whole story. A model with low predictive accuracy can still be financially valuable. The critical factor is not the precise point forecast, but whether the predicted direction or relative changes in the spread series contain useful information that correlates with future price movements.

Therefore, the output of these models should not be dismissed. The next logical step is to transform the predicted sequences into actionable trading rules—for instance, generating long/short signals based on the forecasted spread changes. The true test of the model's utility lies in the performance of the resulting trading strategy, measured by its return series, Sharpe ratio, and maximum drawdown, rather than its fitting precision. When projecting regression results into trading strategies, scores are first assigned upper and lower quantile thresholds within n-day windows (e.g., 0.8/0.2) to generate long/short/neutral signals. Subsequently, signal holding mechanisms (e.g., 5, 10, 20 days) are applied to smooth trading frequency and reduce overtrading risks. Additionally, leverage and margin are simulated in backtests, with maximum capital utilization limits set (e.g., 30%-50%) to prevent liquidation and ensure strategy robustness across different market conditions.

Comparative Analysis of Feature Importance Across Three Strategies

By comparing feature importance analyses in Figures 1, 8, and 15, we can identify significant differences in feature dependencies across different commodity spread strategies. The coking coal-coke strategy heavily relies on fundamental data (inventory, warehouse receipts), reflecting the close connection between ferrous commodities and the physical industry. The crack spread strategy depends more on macroeconomic factors (oil price volatility, US Dollar Index), embodying the financialization characteristics of energy markets. The soybean-palm oil spread strategy shows strong dependence on inventory data, consistent with the cyclical supply-demand patterns of agricultural commodities. These differences provide important references for strategy selection under different market conditions.

Visual Comparison of Strategy Performance

Comparing the equity curves in Figures 2, 9, and 16 clearly reveals performance differences among the three strategies: the coking coal-coke strategy exhibits the steepest and most stable equity curve; the soybean-palm oil spread strategy performs strongly in later periods; while the crack spread strategy shows relatively modest growth with high volatility. These differences stem from the structural characteristics of different commodity markets—the mean reversion properties of ferrous energy commodity spreads are most stable, soybean-palm oil spreads rank second, while the high volatility and frequent extreme events in crude oil markets increase arbitrage difficulty.

Systematic Comparison of Risk-Return Characteristics

By comparing return distributions (Figures 5, 12, 19) and rolling performance metrics (Figures 6, 13, 20) across the three strategies, we find: the coking coal-coke strategy has the most symmetric return distribution with controllable tail risk, maintaining consistently high Sharpe ratios; the crack spread strategy exhibits pronounced negative skewness and fat tail characteristics with highly volatile Sharpe ratios; the soybean-palm oil spread strategy falls

between the two but performs excellently under favorable market conditions. These findings validate the intrinsic risk-return characteristics of different commodity spreads, providing a basis for portfolio allocation.

The following tables contain strategy basic information, signal statistics, backtesting parameters, and performance metrics. Subsequently, we provide a comprehensive assessment of strategy performance.

Comprehensive Comparison of Strategy Backtesting Results

Table 1: Strategy Basic Information and Backtesting Parameters

Indicator	Coking Coal-Coke Arbitrage	Crack Spread	Soybean-Palm Oil Spread
Data Points	583	1242	599
Initial Capital	1,000,000	1,000,000	1,000,000
Commission Rate	0.050%	0.050%	0.050%
Slippage Rate	0.010%	0.010%	0.010%
Leverage	1x	1x	1x
Margin Ratio	10.0%	10.0%	10.0%
Max Capital Utilization	30%	50%	30%
Signal Holding Days	10 days	20 days	20 days

Table 2: Signal Distribution Statistics (After Signal Holding Applied)

Signal Type	Coking Coal-Coke Arbitrage	Crack Spread	Soybean-Palm Oil Spread
Long Signals	256	603	226
Short Signals	231	576	331
Neutral Signals	96	63	42
Long Percentage	43.9%	48.6%	37.8%
Short Percentage	39.6%	46.4%	55.4%
Neutral Percentage	16.5%	5.1%	7.0%

Table 3: Performance Metrics Comparison

Performance Metric	Coking Coal-Coke Arbitrage	Crack Spread	Soybean-Palm Oil Spread
Total Return	154.49%	31.50%	97.28%
Annualized Return	49.39%	5.81%	32.84%
Annualized Volatility	38.12%	24.47%	45.18%
Sharpe Ratio	1.2430	0.1556	0.6826
Sortino Ratio	1.5586	0.2204	1.0690
Calmar Ratio	1.8692	0.1480	1.1218
Maximum Drawdown	-26.42%	-39.23%	-29.27%
VaR (95%)	-1.92%	-2.20%	-1.43%
CVaR (95%)	-4.27%	-3.41%	-4.87%
Win Rate	55.76%	38.35%	54.92%
Profit/Loss Ratio	2.60	1.25	1.54
Total Trades	473	1167	533
Total Commission	9,349.23	22,799.06	12,181.39

This study conducts a systematic backtesting analysis of three futures spread arbitrage strategies—coking coal-coke arbitrage, crack spread, and soybean-palm oil spread—aiming to evaluate their performance and stability under different market conditions. All strategies employ rolling quantile thresholds (window=250, upper quantile=0.8, lower quantile=0.2) to generate trading signals and introduce signal holding mechanisms to smooth trading frequency.

From the performance results, the coking coal-coke arbitrage strategy stands out most prominently, achieving a total return of 154.49%, an annualized return of 49.39%, and a Sharpe ratio of 1.24, significantly higher than the other two strategies. Its risk control capability is also strong, with a maximum drawdown of -26.42%, and both VaR and CVaR within controllable ranges. The strategy has a win rate of 55.76% and a profit-loss ratio of 2.60, indicating high profit quality and risk-adjusted returns.

The soybean-palm oil spread strategy ranks second, with a total return of 97.28% and an annualized return of 32.84%, but exhibits higher volatility (45.18%) and a Sharpe ratio of 0.68. Although its maximum drawdown of -29.27% is slightly higher than the coking coal-coke strategy, its Sortino ratio of 1.07 demonstrates certain advantages in controlling downside risk.

The crack spread strategy shows relatively weak performance, with a total return of 31.50%, an annualized return of only 5.81%, and a Sharpe ratio as low as 0.16. Its maximum drawdown reaches -39.23%, with a win rate of only 38.35%, indicating that the strategy failed to effectively capture profit opportunities amid market volatility, and high trading frequency (1,167 trades) led to substantial commission expenses, further eroding profits.

In summary, the coking coal-coke arbitrage strategy performs optimally in terms of profitability, risk control, and trading costs, making it suitable as a core allocation. While the soybean-palm oil spread strategy offers attractive returns, its volatility risk requires attention. The crack spread strategy requires further optimization of signal generation and filtering mechanisms to enhance its practical applicability. The results of this study provide empirical references for the construction and selection of futures spread arbitrage strategies.

Discussion

The main findings of this study focus on the following key dimensions:

Strategy Adaptability and Market Conditions: By comparing the performance of the three strategies across different market cycles (see monthly return heatmaps in Figures 4, 11, and 18), we find that the coking coal-coke strategy

maintains relatively stable performance across various market environments, while the crack spread strategy is highly sensitive to extreme events, and the soybean-palm oil spread strategy performs well under specific supply-demand configurations. This indicates that strategy selection should fully consider the market structural characteristics of underlying commodities.

Effectiveness of Feature Engineering: Feature importance analysis (Figures 1, 8, and 15) reveals core factors driving spread dynamics across different markets. Fundamental features (inventory, warehouse receipts) contribute most to the coking coal-coke strategy, validating the predictive value of physical industry data for ferrous commodity spreads. The importance of macroeconomic factors in crack spreads highlights the global interconnectedness of energy markets. The significant influence of seasonal factors on soybean-palm oil spreads reflects the cyclical patterns of agricultural commodities. These findings provide direction for targeted optimization of feature engineering.

Necessity of Risk Control Mechanisms: Trade analysis charts (Figures 7, 14, and 21) show that the balance between win rate and profit-loss ratio is crucial for long-term strategy performance. The crack spread strategy's low win rate (38.35%) and high trading frequency (1,167 trades) lead to substantial profit erosion from transaction costs, while the coking coal-coke strategy's high profit-loss ratio (2.60) effectively amplifies gains from favorable trades. This suggests the need to introduce more rigorous signal filtering and position management mechanisms in strategy design.

Challenges in Model Robustness: Rolling performance metrics (Figures 6, 13, and 20) reveal stability differences across different periods. The crack spread strategy's Sharpe ratio deteriorates sharply during high volatility periods, indicating insufficient generalization capability for extreme situations outside the training period. This highlights the need to cover multiple complete market cycles in sample selection and introduce more indicators characterizing extreme states in feature engineering.

Research Limitations: This study has the following limitations: (1) Backtests do not fully simulate microstructure trading costs such as slippage and market impact, so live trading performance may be lower than backtest results; (2) Limited data sample duration, particularly the crack spread strategy failing to adequately cover complete energy price cycles; (3) The model assumes relatively stable statistical properties of spreads, which may fail during structural market changes (e.g., policy adjustments, capacity transformations); (4) This study uses fixed quantile thresholds to generate signals, without considering dynamic adjustment based on market states.

Extended Exploration: Building an LLM-Enhanced News-Driven Risk Management Tool

As demonstrated in the preceding analysis, quantitative models based on historical data and statistical features exhibit significant limitations when confronted with unpredictable black swan events—such as geopolitical conflicts, sudden policy shifts, pandemic outbreaks, or extreme weather conditions. Traditional models, trained exclusively on historical patterns, struggle to anticipate and respond to unprecedented market disruptions that lack historical precedent in the training data.

These structural limitations stem from the fundamental nature of statistical learning: models extrapolate based on observed patterns but inherently lack the capacity to incorporate real-time qualitative information about emerging risks. Political decisions, regulatory changes, international sanctions, or supply chain disruptions often manifest first in news flows and public discourse before materializing in market prices. By the time these events are reflected in quantitative indicators, substantial losses may have already occurred.

To address this critical gap, we introduce a Large Language Model (LLM)-enhanced news-driven risk management system as a complementary layer to our quantitative trading framework. This system leverages the natural language understanding capabilities of modern LLMs to continuously monitor, analyze, and interpret news streams from multiple sources—including financial news agencies, policy announcements, industry reports, and social media sentiment. The LLM serves as an early warning mechanism, identifying potential market-moving events and assessing their relevance and potential impact on specific commodity spreads before they fully propagate through market prices.

System Architecture and Workflow Design

The Large Language Model (LLM)-enhanced news-driven risk management system is realized with workflow. The original intent of this workflow is to provide an intelligent risk management mechanism for spread trading strategies. Any quantitative strategy—whether based on statistical arbitrage, spread models, or cointegration—ultimately relies on relatively stable fundamental assumptions. For example, the coking coal–coke spread depends on supply–demand balance and coking margin logic, while the soybean oil–palm oil spread depends on climate, policy, and trade flows. If these fundamentals undergo abrupt and significant changes—such as policy adjustments, capacity outages, extreme weather, or export restrictions—the relationships that the strategy relies on may break down.

This news-based risk management tool is designed to act precisely in such circumstances. It automatically monitors, on a daily basis, news and policy changes related to the fundamentals of the relevant instruments, analyzes the potential impact of these events by using LLM, evaluates whether there is a risk that could invalidate the strategy, and, when necessary, issues a pause signal. In this way, the strategy system not only gains self-correction capability but can also, with human oversight when needed, maintain stable operations amid sudden risk.

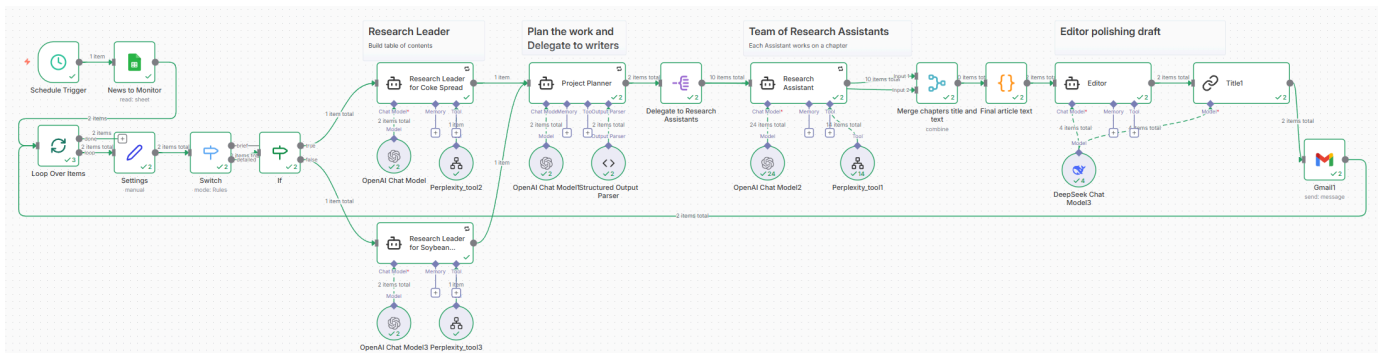


Figure 22: LLM-Enhanced News-Driven Risk Management Workflow Architecture

Figure 22 illustrates the complete workflow architecture of the LLM-enhanced news-driven risk management system. The diagram shows the integrated pipeline from automated news collection through intelligent analysis to decision output, highlighting the collaborative roles of Perplexity API for information retrieval and OpenAI models for semantic analysis and decision-making.

Information Collection and Filtering

The workflow automatically gathers the latest news on the coking coal, coke, soybean oil, and palm oil markets via Perplexity nodes. This layer is the information intake: it runs on a schedule every day, extracting from the web news related to policy shifts, production disruptions, weather disasters, import restrictions, inventory changes, demand expectations, and etc. All information is converted into structured JSON data that includes the date, source, title, facts, and tags indicating potential impacts on the spread.

Intelligent Assessment and Thematic Aggregation

Within the entire system, news collection and analysis are collaboratively executed by the Perplexity API and the OpenAI family of large language models; together they form the workflow's "intelligent information intake layer" and "analytical decision layer."

News Collection Stage: The workflow is triggered by a scheduler at fixed times each day to run the collection tasks automatically. It reads the monitoring topics configured in Google Sheets—such as "China's coking coal–coke futures market" or "China's soybean oil–palm oil spread." These topics are passed to the Perplexity node as query parameters. Perplexity functions like an intelligent search engine: it retrieves from the internet the latest news, announcements, policy documents, or reports by reputable media related to the topic. Unlike traditional web crawlers, Perplexity uses semantic understanding to turn fuzzy search requests into structured fact retrieval. For example, when provided "Recent developments affecting China's coking coal and coke futures markets," the system automatically extracts the date, title, source, URL, and event summary from multiple outlets (including Xinhua, Reuters, Cailian Press, Bloomberg,

Securities Times, etc.). The results are organized into a standardized JSON format in which each item contains a factual abstract, thematic tags, and the original link. This step ensures timeliness and credibility while providing clean inputs for subsequent natural-language analysis.

News Analysis Stage: This stage is completed in sequence by multiple OpenAI-compatible model nodes (including Research Leader, Project Planner, Research Assistant, and Editor). Through internal logic, they call one another to transform raw news into an analytical report that directly supports strategy decisions.

- **Research Leader Node:** reads the output from Perplexity, classifies the facts, and identifies key variables that may affect the spread structure. For the coking coal-coke market, for instance, it emphasizes coal mine production, blast furnace operating rates, policy restrictions, import clearance, and carbon-emission management; for the vegetable-oil market, it focuses on weather conditions, export taxes, import tariffs, biodiesel mandates, and crushing margins. Based on these topics, the Research Leader creates an analytical framework that clarifies which dimensions subsequent analysis should cover.
- **Project Planner Node:** takes this framework and converts it into a complete writing plan. It breaks the research content into multiple chapters—such as policy environment, supply-demand dynamics, inventories and production, weather and logistics, and risk triggers. Each chapter is assigned a specific writing prompt instructing the downstream Research Assistant model what facts to retrieve and what questions to answer in that section.
- **Research Assistant Node:** calls the Perplexity tool again for each chapter to conduct more granular secondary retrieval, obtaining the most relevant and up-to-date information for the chapter's theme. It produces paragraphs with citations in every chapter to ensure that each fact comes from a verifiable source. The output is saved in HTML format and includes data, conclusions, and the news references.
- **Editor Node:** All chapters are consolidated and handed to the Editor node. The Editor uses an OpenAI model for semantic consolidation, logical refinement, and structural optimization, and—based on all compiled facts—generates a "Decision Block." The task of this module is to determine whether the strategy should continue to run or be paused. The Editor assigns a confidence score by assessing event severity, the number of independent sources, recency, and whether the event constitutes a systemic shock (for example, nationwide safety crackdowns, export bans, or extreme weather). If the evidence is insufficient, the default output is "Continue"; only when the evidence is clear and severe does it output "Pause."

In essence, the entire process implements an automated pipeline "from the open web to strategy decisions." Perplexity is responsible for data acquisition and factual structuring, ensuring broad coverage and strong timeliness; the OpenAI models perform the reasoning from facts to logic and from logic to strategy, generating the final text. Combined, they form an adaptive risk-monitoring system that delivers automatically generated, fully sourced, confidence-scored reports to the strategy team every day, thereby enabling fundamentals-driven risk control without manual intervention.

Email and Visual Output

After the report is generated, the Title node automatically extracts key information, produces a subject line that includes "Gate Decision + Confidence," and sends the email. A typical subject might be: "[Pause] — Coking-Coal Spread — Demand Outpacing Supply — 85%." Upon receiving it, traders can quickly determine whether the automated trading system should be paused to mitigate potential risks.

Examples of email received by the trader for Coking-Coal Spreads: <https://drive.google.com/file/d/1QS1gm-x9bnB2xUCkY97VxmFvE4Dwr07q/view?usp=sharing>

Examples of email received by the trader for Soybean Spreads:
https://drive.google.com/file/d/1adBLQs7IbhnolUJIJ3yrrTbJ_o7u099U/view?usp=sharing

AI Declaration

The report was obtained after we listed the detailed outline and handed it over to the ai, and then deleted the non-compliant content generated by the ai